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Dynamical causality under latent confounders for biological network reconstruction

Jinling Yan[†], Shao-Wu Zhang^{*}, Chihao Zhang[†], Weitian Huang, Jifan Shi, Luonan Chen^{*}

Abstract—Causal interaction inference is prone to spurious causal interactions, due to the substantial confounders in a biological system. While many existing methods attempt to address misidentification challenges, there remains a notable lack of effective methods to infer causal interaction under latent/unobserved confounders. In this work, we propose a method to overcome such challenges to infer dynamical causality under latent confounders and further reconstruct the latent confounders from time-series data by developing an orthogonal decomposition theorem in a delay embedding space. This theoretical foundation ensures the causal detection for any high-dimensional system even with only two observed variables under many latent confounders, which is a long-standing problem in the field. In addition to the latent confounder problem, such a decomposition makes the coupled variables separable in the embedding space, thus also solving the non-separability problem of causal inference. Extensive validation of the CIC method is carried out using various real datasets, which all demonstrates its effectiveness to reconstruct real biological networks and unobserved confounders.

Index Terms—Causal inference, orthogonal decomposition, confounder reconstruction, time series, network reconstruction

1. INTRODUCTION

Reconstructing biological network is a fundamental challenge in biology that aims to reveal the topological relationships between components like genes, proteins, and metabolites. Deciphering these networks is crucial for uncovering the important regulatory mechanisms underlying vital life processes and disease pathogenesis [1, 2]. However, inference of such networks is profoundly complicated by the intrinsic complexity and latent confounding present in biological systems.

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Causal inference provides a solution to above challenges, as it seeks to detect directed, functional relationships within naturally evolving systems. Inferring causal interactions in complex systems is crucial yet challenging across various disciplines [3, 4], including economics [5], biology [6], ecology [7] and deep learning [8, 9], especially when latent/unobservable confounders distort causal signals. For example, within gene regulatory networks, unmeasured genetic factors or environmental variability can act as latent confounders, inducing spurious associations between genes [10, 11]. Similar issues arise in ecological and economic dynamics, where climatic shifts in ecosystems or policy-induced shocks in markets generate coherent fluctuations across observed variables. Across these domains, such unobservable confounders cannot be directly measured but nevertheless influence observed outcomes, leading to biased estimations and even incorrect causal directions [12]. Therefore, a fundamental challenge emerges — one that underpins our methodological framework: disentangling true causal interactions from spurious associations induced by latent confounders, while further reconstructing these latent variables themselves. A straightforward solution is to include both the variables, e.g., biomolecules under study and all potential confounders into the observations. However, in practice, many are technically difficult or impossible to observe. Consequently, developing an accurate and reliable data-driven causal detection method that is robust to latent confounders becomes imperative. This has been a long-standing challenge in the fields of data science and deep learning.

Depending on the type of the measured data, there are mainly two frameworks for causal interaction inference: statistical approaches for cross-sectional data and dynamical approaches for time-series analysis [13, 14]. Classical statistical

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The datasets used in this work and the source code for CIC is available at <https://github.com/JinlingY/CIC>.

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frameworks, such as Rubin's potential outcome model (POM) [15] and Pearl's structural causal model (SCM) [16-18], rely on randomized interventions or explicit directed acyclic graph (DAG) assumptions (e.g., Peter-Clark (PC) algorithm [19], LiNGAM [20], HSIC-NNM [21], Differentiated Confounder Balancing [22]). These methods excel in linear or parametrized systems with static data, while they critically assume underlying system adheres to a DAG structure or no unobserved confounders, rendering them ineffective for complex nonlinear dynamical systems, and vulnerable to unobserved confounders.

To overcome these limitations, researchers proposed many effective algorithms from a dynamical perspective to infer causal interaction for time-series data [23], including Granger causality (GC) [24], mutual prediction method [25], state space method [26], quantifying information method [27], convergent cross method (CCM)[4], cross map evaluation(CME) [28], cross map smoothness (CMS) [29], partial cross mapping (PCM) [3] and conditional cross-map-based techniques [14, 30, 31], etc. GC is a widely used predictability-based measure of dynamical interactions. However, GC is limited to linear causality. Transfer entropy (TE) and Neural Granger causality [32] extend GC to nonlinear cases, but they still suffer from the non-separability problem of nonlinear dynamics, i.e., the components within one system intertwine with each other, and the dynamics of each component cannot be separated from one another. Recurrent model-based Granger Causality [33] utilizes the feed-forward neural networks to infer representations of the confounders from available proxy variables. InvarGC [34] leverages cross-environment heterogeneity to mitigate the effects of latent confounder, thereby recovering invariant causal relations. These methods extend the capability of Granger causality to handle latent confounders, but fundamentally rely on strong auxiliary assumptions, i.e., learned proxy representations or sufficient cross-environment invariance, and cannot truly quantify latent confounders. Methods like CCM, CME, CMS, rooted in Takens' embedding theory and state space reconstruction technique, excel in time-dependent systems or nonlinear dynamical systems. And the erroneous or spurious causal interaction caused by visible/observable variables can be eliminated by conditional causal methods, such as PCM, conditional Granger causality (cGC). However, these methods need to iterate through all observed variables to exclude spurious or false causality, leading to high computational cost and instability. Therefore, challenges arise when attempting to detect causality with latent confounders.

Recent advances in causal interaction inference aim to combine the statistical causal frameworks or deep learning techniques with dynamical processes [35] to address these issues, e.g., Peter-Clark momentary conditional independence (PCMCI) [36], DYNOTEARS[37], Double Machine Learning (DML) [38], the method based linear NDS[39] and Causal Mediation Analysis Variational Auto-Encoder (CMAVAE)[40]. These methods allow for wide applications of causal inference in dynamic scenarios. PCMCI [36], DYNOTEARS [37] and DML [38] are dependent on the assumption of Causal Sufficiency, thus without addressing the challenges posed by latent confounders. The method in [39] dedicates to deal with spurious associations generated by structured noise. And the noise in proxy variables and their weak correlations with

unobserved confounders may inherently lead to residual confounding effects for CMAVAE [40]. In summary, most methods struggle to infer causal interactions in the presence of confounders, some of which can handle observed confounders, but the problem of accurately inferring causal interactions with latent confounders for a general system, particularly in nonlinear dynamical systems, is still an unresolved challenge.

We summarize the crucial problems of causal inference under latent confounders as follows:

- (1) *Nonlinear dynamic scenario*: most real-world systems are nonlinear, complex, and temporally dependent. Traditional causal assumptions—such as the Causal Markov Assumption and Faithfulness Assumption, limited the advancements of causal inference research in the field of nonlinear dynamical systems.
- (2) *Non-separability*: the coupled interactions among the components of a nonlinear dynamical system make it difficult to separate latent confounders from the observed time series data.
- (3) *The spurious causality caused by latent confounders*: how to distinguish true causal interactions from spurious confounding effects, and how to reconstruct unobserved confounder from observed time series data are very interesting and important problems.

To fill this gap, we develop Causality under Latent Confounders (CIC), a computational framework with theoretical advancements for detecting dynamical causality from time series data. CIC accurately detects causal associations and further reconstructs latent confounders, only from the observed data. The core idea is to transform original variables into delay-embedded forms, decompose them into common and private subspaces, and thereby quantify causal relationships among them. Our novel orthogonal decomposition theorem ensures the separability of coupled variables in the embedding space, resolving the non-separability issue caused by latent confounders. Note that effect variables reconstruct their causal variables in the delay embedding space in contrast to the original space where cause variables predict their effect variables. Computationally, we employ deep neural networks to perform the orthogonal decomposition effectively, enabling accurate causality detection from observed time-series data even with unobserved confounders. Our framework rests on the theoretical basis of the orthogonal decomposition theorem for dynamical causality, while deep learning computationally realizes nonlinear variable decomposition and latent confounder reconstruction.

The main contributions of our work are as follows,

- The CIC method elegantly addresses the long-standing problem of misidentifying spurious causal influence due to confounders, even with unobserved confounders, using only two observed variables. It stands out from existing causal inference methods by effectively reconstructing latent confounders.
- The orthogonal decomposition theorem ensures the decomposition in a delay-embedded space, making the coupled variables separable in the embedding space and thus solving the non-separability problem of latent confounders.
- By the delay embedding transformation and orthogonal

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decomposition, the CIC method reduces reliance on traditional causal assumption, thus enhancing causal inference in nonlinear dynamical systems.

- The method is extensively validated through various benchmark systems and real datasets with different causal structures. The applications to various biological systems demonstrate its effectiveness as a powerful tool to analyze and reconstruct the real biological networks, solely using time series data of partially observed variables.

2. RELATED WORKS

A. Statistical causality inference

Classical statistical frameworks—including potential outcome model (POM) [15] and structural causal model (SCM) [16-18]. According to how confounders are addressed, the existing causal discovery methods can be divided into the following categories: (1) Constraint-based algorithms; (2) Score-based algorithms; (3) Functional-based algorithms; (4) Continuous Optimization-based algorithms; (5) Prior Knowledge-based and Hybrid Methods. Representative methods of constraint-based algorithms, e.g., FCI [41], and its variant (e.g., RFCI) [42], can address latent confounders by producing a partial ancestral graph (PAG) [43] identifiable under the Markov and faithfulness assumptions within the Maximal Ancestral Graph (MAG) framework, but cannot fully orient edges or recover a unique DAG. Conditional independencies-based method [44] transforms the identification of latent confounders into statistically testable hypotheses by leveraging multi-environment observational data, but it relies entirely on access to multiple-environment datasets and strong independence assumptions. RelFCI [45] integrates the FCI and Relational Causal Discovery (RCD) algorithms via newly introduced relational graphical models, enabling causal discovery with hidden variables in relational data. Score-based algorithms, e.g., GES [46], RL-BLC [47], typically assume causal sufficiency, and thus inapplicable in the presence of latent confounders. Extensions such as LRpS-GES [48] uses the low-rank plus sparse estimator to remove confounding before applying GES, but relies on linear Gaussian models assumptions and thus is not robust in nonlinear non-Gaussian settings. The functional-based algorithms, e.g., OICA, lvLiNGAM and DeFuSE, aim to model functional relations directly. OICA [49] identifies the latent variables and determine causal orders by estimating the mixing matrix using the higher-order cumulants. However, OICA relies heavily on the assumed latent-component structure, and its performance can degrade substantially when the confounding pattern is more complex or nonlinear. lvLiNGAM [50], as an extension of LiNGAM, employs OICA to estimate the mixing matrix from observed data and then recover the causal structure. DeFuSE [51] learns functional causal relations while adjusting for linear confounding effects, however, its ability to handle nonlinear confounders remains limited. The Continuous Optimization-based algorithms recast discrete DAG search as continuous optimization problem, notably NOTEARS [52] and DAG-GNN [53]. These methods achieve efficient computation and scalability, and can adjust for observed confounders, but provide no inherent mechanism for latent confounders. Prior Knowledge-based and Hybrid Methods, e.g., JCI algorithm [54], incorporate auxiliary context

variables across multiple environments, enabling causal discovery even in the presence of latent confounders and interventions, but its reliance on faithfulness assumptions regarding context variables may limit applicability in complex designs. In summary, these methods excel in time-independent or parametrized systems with static data, while their reliance on strong assumptions (e.g., Causal Markov and Causal Sufficiency) limits their effectiveness in complex nonlinear dynamical systems and makes them vulnerable to latent confounders.

B. Dynamical causality inference

To overcome the above difficulties, many causal inference algorithms from the dynamical perspective [23] were proposed, including GC [24], TE, CCM [4], PCM [3], EE [13], perturbation cascade inference (PCI) [55], General ODE-Based Inference (GOBI) [56] and Information Imbalance [23]. GC, TE and their variants have been continuously used to infer causal associations from time-series data [43, 57], but they still suffer from the non-separability problem. CCM can deal with the system with interdependence, e.g., ecological system, where GC is simply not applicable, but CCM usually struggles to distinguish between direct and indirect causation. To prevent such false positive predictions, PCM was proposed to eliminate indirect causal interactions. However, the local linear operations in CCM/PCM limit their performance in detecting nonlinear causality. EE in [13] first proposes the dynamical causality (DC) framework, which provides mathematical insights into causality. PCI is an active inference approach, which infers the structure of a dynamical network through systematic perturbations of its components, while it requires the user to sort nodes in the order in which they become perturbed using a system-specific time series analysis. GOBI can be applicable to the monotonic system with positive and negative regulations described by ODE, which can also distinguish between direct and indirect causation, however, the effect of unobserved confounders is not further discussed. Information Imbalance, as is a nonlinear and model-free generalization of GC, detects causal interaction in time-ordered data based on a statistical measure constructed on distance ranks, but still does not address the problem of unobserved confounders.

C. Combined causal framework

Recent advances in causal interaction inference aim to combine the statistical causal frameworks or deep learning techniques with dynamic analysis [35] to address these issues, e.g., BETS [58], SWING [59] G1DBN [60], PCMCI [36], DYNOTEARS[37], DML[38], Intervened reservoir computing (IRC) [61] and CMAVAE[40]. BETS, a statistical framework based on GC was proposed to recover a directed gene network from transcriptional time-series data. SWING improves upon existing Granger methods by generating an ensemble of windowed models that simultaneously evaluate multiple upstream regulators at several potential time delays. G1DBN is an inference method for dynamic genetic networks which makes it possible to face a number of time measurements, smaller than the number of genes based on the theory of graphical models associated with the directed acyclic graphs (DAGs). BETS, SWING and G1DBN have been used to infer

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gene regulation network based on the time-series data, but cannot deal with the influence of unobserved factors, such as chromatin structure and epigenetic regulations. PCMCi combines linear or nonlinear conditional independence tests with a causal discovery algorithm to estimate causal networks from large-scale time series datasets. DYNOTEARS as a dynamic Bayesian networks method, formulates structure learning of DAG as an optimization problem. DML reduces bias from observed confounders by orthogonalizing treatment and outcome variables. However, PCMCi, DYNOTEARS and DML are inevitably dependent on the assumption of Causal Sufficiency, thus without addressing the challenges posed by latent confounders. Another novel method based on machine learning was proposed to learn the hidden causal network of a linear networked dynamical system (NDS) from the time series data based on partial observability, which dedicates to deal with spurious associations generated by structured noise. IRC [61] can detect causal interactions by constructing a neural network replica of the original system and applying interventions to it, which provides valuable information regarding the underlying causal strengths but cannot detect hidden nodes or perform causal discovery with the existence of latent confounders. CMAVAE method incorporates proxy variables and deep latent-variable models to estimate both the causal effects and the confounders, and overcomes the reliance on the sequential ignorability assumption in traditional CMA methods. But the noise in proxy variables and their weak correlations with unobserved confounders may inherently lead to residual confounding effects. The LPCMCi method [62], an extension of PCMCi, achieves high recall in detecting latent confounders by identifying bidirectional links in the causal graph. However, it provides only qualitative evidence of latent confounder and is limited to non-autocorrelated time-series data and some nonlinear cases. Therefore, challenges arise when attempting to detect causal interaction in nonlinear dynamical system with latent confounders.

3. PROBLEM DEFINITION

A biological system is generally composed of a large number of components, e.g., biomolecules or species, many of which are unobservable but affect the causal inference between variables, known as latent confounders. Assume that time series (length L) of two variables x_t and y_t , e.g., $\{x_1, \dots, x_t, \dots, x_L\}$ and $\{y_1, \dots, y_t, \dots, y_L\}$ are observed in a dynamical biological system, while all other variables in the system are not observed. The causality in the original space is defined as follows: if $y_t = f(x_{t-1}, y_{t-1})$ and $x_t = g(x_{t-1}, y_{t-1})$, then $x \rightarrow y$. However, a case that will be misidentified as causality is when $y_t = f(y_{t-1}, z_{t-1})$ and $x_t = g(x_{t-1}, z_{t-1})$, where z_{t-1} is an latent confounder of x_t and y_t , e.g., nearby variants and genetic components of other genes' expression. Then only with observed data of x and y , it is difficult to infer correct causality $x \rightarrow y$ in the original space due to their latent confounder set z and the non-separability of x_t and y_t .

This brings up two challenges (Fig. 1A) for causality detection from observed data:

1) How to reliably infer the causal interaction from x to y in the presence of latent confounders?

2) How to reconstruct those latent confounders from the observed time series?

4. PROPOSED METHOD

4.1 Overview of CIC method

To solve such problems, a new framework based on orthogonal decomposition theorem in a delay embedding space is proposed to infer causal interactions even under latent confounders.

1. **Delay embedding transformation:** the theoretical basis (Supplementary Materials 1.1) of this method is the delay embedding theorem. Takens' delay embedding theory ensures that the delay coordinate $[x_i(t), x_i(t - \tau), \dots, x_i(t - (L - 1)\tau)]$ of a variable x_i is a state space reconstruction of the original system related to this variable. The Takens' delay embedding theory is introduced to transform the original data x and y into the embedding data X and Y , where $X_{t-1} = [x_{t-1}, x_{t-2}, \dots, x_{t-p}]^T$, $Y_t = [y_t, y_{t-1}, \dots, y_{t-(p-1)}]^T$, respectively, i.e. $X_{t-1}, Y_t \in R^p$ (Fig. 1B), i.e. from original space to delay embedding space.
2. **Orthogonal decomposition for X_{t-1} and Y_t .** We develop the Orthogonal Decomposition Theorem (Fig. 1A) in delay embedding space and propose the computational method of orthogonal decomposition based on VAE (Fig. 1C). Here the deep learning techniques are introduced as the backbone to computationally realize such information decomposition of variables. Then the embedding space X_{t-1} or Y_t is theoretically and computationally ensured to decompose into two orthogonal subspaces at each instant t respectively, i.e., the common space/information z_{xy} between X_{t-1} and Y_t , and the private space/information z_x (or z_y). Thus, we can construct the causal criterion under latent confounders, i.e. $CIC_{x \rightarrow y}$ (Fig. 1D).
3. **Causal interaction inference:** Then, according to our orthogonal decomposition theorem, if $CIC_{x \rightarrow y} \rightarrow 1$ then $x \rightarrow y$. Otherwise, if $CIC_{x \rightarrow y} \rightarrow 0$, then $x \nrightarrow y$; especially, if $CIC_{x \rightarrow y} \in (0, 1)$ and $y \nrightarrow x$, there is a confounder between x and y , and the latent confounder can further be quantified/reconstructed by the shared latent subspace z_{xy} . Thus, the causal criterion $CIC_{x \rightarrow y}$ can make causal interaction inference under latent confounders (Fig. 1D), and the latent confounders can also be quantified/reconstructed by the shared latent subspace.

In summary, our CIC serves two key functions: firstly, inferring the causal interaction between pairwise variables even from the observed data of only these two variables, and secondly, reconstructing the latent confounders of pairwise variables.

4.2 Assumptions of CIC method

The assumptions of our CIC framework are temporal precedence (e.g., $x_{t-1} \rightarrow y_t$) for a deterministic system with the observed states on a compact manifold. The details about assumptions and the differences of CIC method from classical causal assumptions are provided in Supplementary Materials 1.2.

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4.3 Definitions of dynamical causality

Consider a general discrete dynamical system with n variables in the form of

$$\mathbf{x}_t = \mathbf{f}(\mathbf{x}_{t-1}), \quad (1.1)$$

with an initial value $\mathbf{x}_0$ and $t = 1, 2, \dots$, where $\mathbf{f} = [f_1, f_2, \dots, f_n]^T$ is a nonlinear function vector, and $\mathbf{x}_t = [x_{1,t}, x_{2,t}, \dots, x_{n,t}]^T$ is the system's state vector which describes the evolution of the system along time. For the case of two observed variables (x, y) shown in **Fig. 1A**, we simply denote as $x_{i,t} = x_t$ and $x_{j,t} = y_t$. Here, one unit time interval is Δt . We first define dynamical causality in the original space.

Definition 1 (Dynamical causality in the original space): We call that x_i has the dynamical causality (DC) to x_j , or the dynamics of x_j depend on x_i , if the following condition of eq. (1.1) holds

$$\frac{\partial f_j(x_{t-1})}{\partial x_{i,t-1}} \neq 0$$

for almost any sampled t , and we denote it as $x_i \rightarrow x_j$. Otherwise, we say that x_i has no dynamical causality to x_j , i.e., $x_i \nrightarrow x_j$ [13, 14].

We next transform the original space to a delay embedding space. For the sake of simplicity in description, we consider a discrete dynamical system of $x \rightarrow y$ with z as a latent variable, as follows

$$\begin{cases} x_t = g(x_{t-1}; z_{t-1}), \\ y_t = f(x_{t-1}, y_{t-1}; z_{t-1}), \end{cases} \quad (1.2)$$

where f and g are two nonlinear functions (continuously differentiable), i.e. $x_t = x_{i,t}$ and $y_t = x_{j,t}$ with $z_t = x_{k,t}$ for eq. (1.1). By combining all p equations of $f(x_{t-1}, y_{t-1}; z_{t-1}) - y_t = 0$ with $t, t-1, \dots, t-(p-1)$ of the second eq. (1.2), we have

$$H(X_{t-1}, Y_t; Z_{t-1}) = 0, \quad (1.3)$$

where $H = [f(x_{t-1}, y_{t-1}; z_{t-1}) - y_t, \dots, f(x_{t-p}, y_{t-p}; z_{t-1}) - y_{t-(p-1)}]^T$ is a nonlinear function vector; $Y_t = (y_t, y_{t-1}, \dots, y_{t-p})^T \in \mathbb{R}^{p+1}$ and $X_{t-1} = (x_{t-1}, x_{t-2}, \dots, x_{t-p})^T \in \mathbb{R}^p$ with $Z_{t-1} = (z_{t-1}, z_{t-2}, \dots, z_{t-p})^T \in \mathbb{R}^p$ are the delay embedding vectors of variables x and y with z , respectively. Clearly, $\frac{\partial f(x_{t-1}, y_{t-1}; z_{t-1})}{\partial x_{t-1}} \neq 0$ for almost any sampled t due to $x \rightarrow y$ or

Definition 1, and thus we have the $p \times p$ diagonal matrix $|\frac{\partial H(X_{t-1}, Y_t; Z_{t-1})}{\partial X_{t-1}}| \neq 0$. Hence, by the implicit function theorem, Eq. (1.3) guarantees the local existence of a smooth reconstruction map

$$X_{t-1} = F'(Y_t; Z_{t-1}), \quad (1.4)$$

However, since Z_{t-1} is not observed, we must rely solely on the observed series Y_t . Furthermore, by Takens' embedding theorem, Y_t provides a diffeomorphic reconstruction of the attractor of system (1.2), hence, the latent state Z_{t-1} is implicitly represented within the coordinates of Y_t . Taken's embedding theorem ensures global existence of the smooth function F of eq. (1.4) such that

$$X_{t-1} = F(Y_t), \quad (1.5)$$

when the embedding dimension $p > 2d$, where d is the

intrinsic dimension of attraction manifold of the dynamics for eq. (1.2) or its box-counting dimension. Generally, d is a small integer for a real system and we can choose an appropriate embedding dimension p . Note, the map F' in Eq. (1.4) arises from the implicit function theorem and is defined locally, depending on both Y_t and Z_{t-1} . In contrast, the map F in Eq. (1.5) follows from Takens' embedding theorem and is defined globally, relying solely on Y_t . Eq. (1.5) allows us to quantify dynamical causality in the embedding space based solely on observations of x and y . Thus, we have the dynamical causality measured in the embedding space as follows. The derivation of eq. (1.5) based on a detailed version with noise term of system eq. (1.2) or eq. (1.1) is provided in **Supplementary Materials 1.3**.

Definition 2 (Dynamical causality based on delay embedding data): we call the dynamical causality (DC) from x to y , i.e. $x \rightarrow y$, if there exists an inverse mapping $X_{t-1} = F(Y_t)$ in the delay embedding space (X, Y) for eq. (1.2).

Note that the cause X_{t-1} is the information at past time $t-1$, while its effect Y_t is the information at current time t . We can reconstruct the past cause by the current effect if $x \rightarrow y$ from the delay embedding data. For a system (1.1) with multiple observed variables, without loss of generality, we can similarly derive the direct dynamical causality $x_i \rightarrow x_j$ on the condition of all other observed variables, yielding the similar result

$$X_{t-1} = F(Y_t; W_{t-1}), \quad (1.5)$$

where $X_{t-1} = (x_{i,t-1}, \dots, x_{i,t-p})^T$, $Y_t = (x_{j,t}, \dots, x_{j,t-(p-1)})^T$, with $W_{t-1} = (x_{k,t-1}, \dots, x_{k,t-(p-1)})^T$ as the delay embedding vector of the all remaining observed variables except x_i and x_j in eq. (1.1). The definition of such direct dynamical causality for eq. (1.1) is given in **Supplementary Materials 1.3** based on eq. (1.5).

4.4 Orthogonal decomposition theorem for dynamical causality

For any pair of observed variables defined in eq. (1.1) or (1.2), e.g. x and y , we assume there exist injective decomposition maps $\mathcal{G}_x, \mathcal{G}_y: \mathbb{R}^p \rightarrow \mathbb{R}^m$ that project the delay-embedding vectors into a common latent space $\mathbb{R}^m$. In that space we assume a direct-sum decomposition into three mutually orthogonal subspaces

$$\mathbb{R}^m = Z_x \oplus Z_y \oplus Z_{xy},$$

denote by

$$\mathcal{G}_x(X_{t-1}) = [z_{x,t-1}; z_{xy,t-1}], \mathcal{G}_y(Y_t) = [z_{y,t}; z_{xy,t-1}],$$

with $z_{x,t-1} \in Z_x$, $z_{y,t} \in Z_y$, $z_{xy,t-1} \in Z_{xy}$. Hence $z_{x,t-1} \perp z_{xy,t-1}$, $z_{y,t} \perp z_{xy,t-1}$, $z_{x,t-1} \perp z_{y,t}$, in which $\perp$ is the symbol indicates orthogonality (i.e., zero inner product in $\mathbb{R}^m$). Because $\mathcal{G}_y$ is injective, Y_t is fully determined by $[z_{y,t}; z_{xy,t-1}]$, i.e. $Y_t \in Z_y \oplus Z_{xy}$. Since any vector in Z_x is orthogonal to $Z_y \oplus Z_{xy}$, it follows that $z_{x,t-1}$ is orthogonal to Y_t , i.e., $z_{x,t-1} \perp Y_t$. Similarly, $z_{y,t} \perp X_{t-1}$. For clarity, we will omit the subscript t or $t-1$ of decomposed variables in the following description. Here z_{xy} also represents the information of X_{t-1} decomposed from Y_t . If z_{xy} contains all the information of X_{t-1} ,

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that is, the private information z_x of X_{t-1} relative to Y_t is trivial, i.e.,

$$\|z_x\| = 0,$$

in which $\|\cdot\|$, measuring the magnitude of each component, denotes the Euclidean norm in the latent space $\mathbb{R}^m$. And thus Y_t contains the complete information about X_{t-1} , combined with the injectivity of $\mathcal{G}_y$, this implies the existence of a reconstruction map F with

$$X_{t-1} = F(z_{xy}) = F(Y_t).$$

and thus, by Definition 2, $x \rightarrow y$. Conversely, if no such F exists, (equivalently $\|z_x\| \neq 0$), then X_{t-1} cannot be recovered from Y_t , and $x \nrightarrow y$. Consequently, according to Definition 2, we have the following theorem for orthogonal decomposition in the latent space.

Theorem 1 (Orthogonal decomposition theorem in delay embedding space): Given time series data for any pair of the observed variables x_t and y_t defined in eq. (1.2), there exist appropriate injective nonlinear decomposition mappings $\mathcal{G}_x: \mathbb{R}^p \rightarrow \mathbb{R}^m$ and $\mathcal{G}_y: \mathbb{R}^p \rightarrow \mathbb{R}^m$ to orthogonally decompose

the embedding vectors X_{t-1} and Y_t , respectively, i.e.

$$\begin{aligned} \mathcal{G}_x(X_{t-1}) &= [z_x; z_{xy}] \text{ with } z_x \perp z_{xy}, \\ \text{and } \mathcal{G}_y(Y_t) &= [z_y; z_{xy}] \text{ with } z_y \perp z_{xy}, \end{aligned} \quad (1.6)$$

and $z_x \perp z_y$, in which $\perp$ is the symbol of orthogonal. Furthermore, we have the following criteria of dynamical causality.

Criterion 1 (Causality): $X_{t-1} = F(z_{xy})$ or $\|z_x\| = 0 \Leftrightarrow x \rightarrow y$;

Criterion 2 (Non-causality): $X_{t-1} \neq F(z_{xy})$ or $\|z_x\| \neq 0 \Leftrightarrow x \nrightarrow y$;

where the function F is the reconstructing function or map from Y_t to X_{t-1} defined in Definition 2.

Remark 1 (Confounder reconstruction): For the case of non-causality from x to y of eq. (1.2), i.e. $X_{t-1} \neq F(z_{xy})$ or $\|z_x\| \neq 0$, we have the following Criterion 3:

Criterion 3 (Confounder): If $\|z_{xy}\| \neq 0$, then there exists a confounder of x and y , which can be quantified by z_{xy} ; Note a critical prerequisite for applying Criterion 3 is the exclusion of reverse causation, i.e., the causal link from y to x .

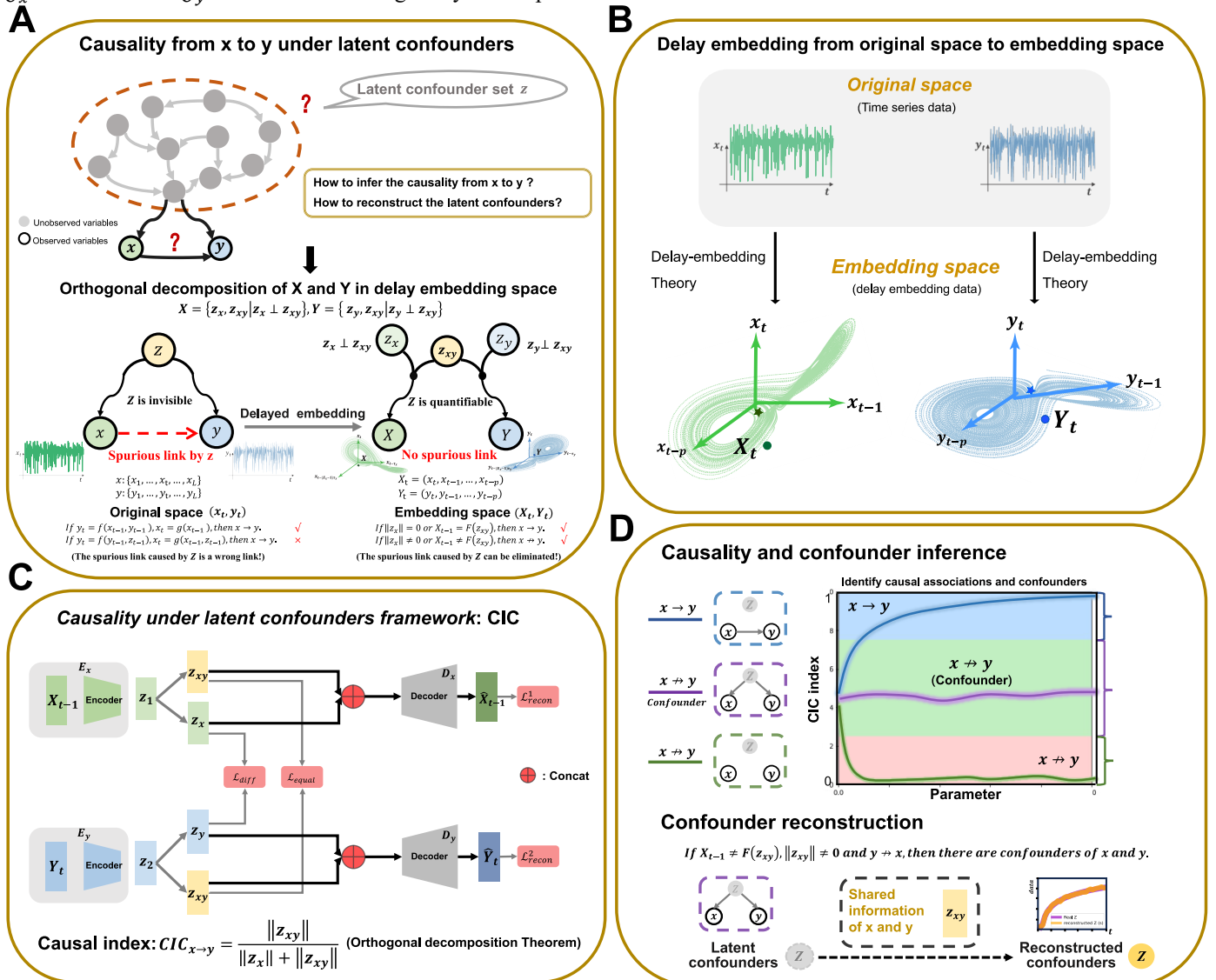


Fig. 1. Framework of the Causality under Latent Confounders (CIC) method. A. Basic principles of CIC. Unobservable variables in dynamical systems complicate causal inference. CIC leverages Orthogonal Decomposition Theorem to eliminate the

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spurious links which are usually mistakenly inferred as causality in the original space. **B. Delay embedding space of x and y .** Time series of original variables x_t and y_t can be transformed into delay embedded X_{t-1} and Y_t . **C. CIC framework.** The delay embedded variable X_{t-1} or Y_t are encoded into two orthogonal latent subspaces, private subspace z_x (or z_y) and common subspace z_{xy} . The decoders reconstruct the inputs through the concatenation of private and common subspaces. $\oplus$: concatenation; $\mathcal{L}_{diff}$ and $\mathcal{L}_{equal}$ are the orthogonality and equivalent constraint, respectively. And the two subspaces $\|z_x\|$ and $\|z_{xy}\|$ are used to construct the causal index $CIC_{x \rightarrow y}$. **D. Detecting the causality and reconstructing latent confounder.** The blue curve, purple curve, and green curve indicate the causality, a confounder, and the non-causality of x and y , respectively. Latent confounder Z of x and y can be quantified through CIC framework.

Remark 2 (Causal index): For any pair of observed variables defined in eq. (1.1) or (1.2), e.g. x and y , we can quantify the dynamical causality from x to y as $CIC_{x \rightarrow y} = \frac{\|z_{xy}\|}{\|z_x\| + \|z_{xy}\|} \in [0,1]$. If $CIC_{x \rightarrow y} \rightarrow 0$, there is no causality from x to y (by $\|z_{xy}\| = 0$ in Criterion 2); if $CIC_{x \rightarrow y} \rightarrow 1$, there is causality from x to y (by Criterion 1); if $CIC_{x \rightarrow y} \in (0,1)$ and $CIC_{y \rightarrow x} \rightarrow 1$ (i.e., $y \rightarrow x$), there is a confounder of x and y (by Criterion 3).

The $CIC_{x \rightarrow y}$ index formalizes causal influence as the state space reconstruction from the effect to the cause, in which $\|z_x\| + \|z_{xy}\|$ represents the total information of X_{t-1} and with $\|z_x\| + \|z_{xy}\| > 0$, while $\|z_{xy}\|$ quantifies the component of X_{t-1} that can be recovered from Y_t . This design translates the conceptual definition of dynamical causality (Definition 2) into a computable, normalized measure. The detailed proof of the three criteria and the Orthogonal Decomposition Theorem under multiple observed variables (to infer direct dynamical causality) are provided in the **Supplementary Materials 1.3**. In particular, Theorem 1 indicates that we can theoretically infer their causal relation using the time-series data of only two observed variables for any high-dimensional system, even if there are possibly many unobserved confounders, which is a long-standing problem in the field. It is noteworthy that the orthogonal decomposition theorem also has limitations when dealing with nonlinear systems. Specifically, it is applicable to nonlinear dynamical systems that satisfy the assumptions of Takens' embedding theorem. The detailed discussion of these limitations is provided in the **Supplementary Materials 1.3**.

4.5 Computational framework for CIC by VAE

To achieve Theorem 1 computationally, we adopt the Variational Auto Encoder (VAE) for nonlinear orthogonal decomposition, allowing causal inference from the observed time series data. The rationale for selecting VAEs to implement orthogonal decomposition is detailed in **Supplementary Materials 1.4**. To infer causal interactions between pairs of variables, e.g. x and y , in a system (1.1) or (1.2), firstly, the original time series data of x_t and y_t are transformed into delay embedding data X_t and Y_t , respectively, according to the Takens' embedding theorem (**Fig. 1B**). The CIC takes the delay embedding vectors X_{t-1} and Y_t as input.

4.5.1 Causal encoder for orthogonal decomposition

Then we make the orthogonal decomposition based on VAE, by decomposing X_{t-1} and Y_t into two orthogonal subspaces at each instant t respectively, i.e., the common space/information z_{xy} between X_{t-1} and Y_t , and the private space/information

z_x (or z_y). The VAE based architectures indeed strive for local orthogonality. Recent works trace the cause of disentanglement in VAEs to the use of diagonal posterior covariance matrices [63-65]. In the inference process, the delay embedded variable X_{t-1} and Y_t are encoded into two latent variables, i.e., z^1 and z^2 , by two different encoders, E_x parametrized by $\emptyset$ and E_y parameterized by $\emptyset'$, respectively,

$$q_{\emptyset}(z^1|X_{t-1}) \sim \mathcal{N}(\mu_1, \sigma_1^2 I),$$

$$q_{\emptyset'}(z^2|Y_t) \sim \mathcal{N}(\mu_2, \sigma_2^2 I).$$

The diagonal posterior covariance matrices of z^1 (or z^2) ensures that the column orthogonality of its Jacobian, which means that its right singular vectors are standard basis vectors that define paths in z^1 (or z^2), over which marginal densities are independent, as a property of $p(z^1)$. Since z_x and z_{xy}^x are derived from different standard basis-vectors of z^1 , i.e.,

$$\mu_1 = \begin{bmatrix} \mu_{z_x} \\ \mu_{z_{xy}^x} \end{bmatrix}, \sigma_1^2 = \begin{bmatrix} \sigma_{z_x}^2 & 0 \\ 0 & \sigma_{z_{xy}^x}^2 \end{bmatrix}.$$

they are conditionally independent given X_{t-1} , i.e.,

$$q_{\emptyset}(z_x, z_{xy}^x|X_{t-1}) = q_{\emptyset}(z_x|X_{t-1})q_{\emptyset}(z_{xy}^x|X_{t-1}).$$

Similarly, z_y and z_{xy}^y derived from z^2 are also conditionally independent given Y_t ,

$$q_{\emptyset'}(z_y, z_{xy}^y|Y_t) = q_{\emptyset'}(z_y|Y_t)q_{\emptyset'}(z_{xy}^y|Y_t).$$

Thus, the disentangled representations of X_{t-1} are sampled from multivariate Gaussians:

$$q_{\emptyset}(z_x|X_{t-1}) \sim \mathcal{N}(\mu_{z_x}, \sigma_{z_x}^2 I), \quad (4.1)$$

$$q_{\emptyset}(z_{xy}^x|X_{t-1}) \sim \mathcal{N}(\mu_{z_{xy}^x}, \sigma_{z_{xy}^x}^2 I). \quad (4.2)$$

Likewise, the disentangled representations of Y_t are sampled from the multivariate Gaussian

$$q_{\emptyset'}(z_y|Y_t) \sim \mathcal{N}(\mu_{z_y}, \sigma_{z_y}^2 I), \quad (4.3)$$

$$q_{\emptyset'}(z_{xy}^y|Y_t) \sim \mathcal{N}(\mu_{z_{xy}^y}, (\sigma_{z_{xy}^y}^2 I)). \quad (4.4)$$

Therefore, due to the column orthogonality of the Jacobian, the latent variables z_x and z_{xy}^x (or z_y and z_{xy}^y) map to independent components in input X_{t-1} (or Y_t) and factorize the density in X_{t-1} (or Y_t), thus to promote disentanglement in VAE [65].

To further enhance orthogonal decomposition of X_{t-1} and Y_t , i.e., z_x orthogonal to z_{xy}^x , z_y orthogonal to z_{xy}^y , and z_x orthogonal to z_y , an orthogonal constraint $\mathcal{L}_{diff}$ based on cosine similarity is introduced, i.e.

$\mathcal{L}_{diff} = ortho(z_x, z_{xy}^x) + ortho(z_y, z_{xy}^y) + ortho(z_x, z_y)$, where $ortho(a, b) = \|\cos(a, b)\|^2$. In addition, another constraint on the latent spaces is also introduced to control the orthogonal decomposition. According to Theorem 1, the sharing information z_{xy}^x from X_{t-1} and z_{xy}^y from Y_t are equivalent, i.e., $z_{xy} = z_{xy}^x = z_{xy}^y$, thus a soft constraint of equivalence is introduced into the computational framework,

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$$\mathcal{L}_{equal} = MSE(z_{xy}^x, z_{xy}^y).$$

4.5.2 Causal decoder for reconstruction and optimization

In the generative process, the concatenation of the common latent vector z_{xy}^x and the private latent vector z_x are expected to reconstruct the original delay embedding vector X_{t-1} through the Decoder D_x parametrized by θ , i.e., $\hat{X}_{t-1}$. And the concatenation of the latent spaces z_{xy}^y and z_y are expected to reconstruct the original delay embedding vector Y_t through the Decoder D_y parametrized by θ' , i.e., $\hat{Y}_t$ (Fig. 1C).

Note that there are also two private/special reconstructions by z_x and z_{xy} , respectively (see Fig. S3). Here, $\hat{X}_{t-1}^{z_{xy}}$ is the reconstruction of X_{t-1} by the common latent vector z_{xy} and an all-zero vector z_x^0 with the same dimension like z_x , which can also be regarded as a reconstruction of X_{t-1} by Y_t , and $\hat{X}_{t-1}^{z_x}$ is the reconstruction of X_{t-1} by the private latent vector z_x and an all-zero vector z_{xy}^0 with the same dimension of z_{xy} .

The variational inference aims to maximize the evidence lower bound (ELBO) by learning the encoder E_x (or E_y) and the decoder D_x (or D_y), i.e.,

$$\mathcal{L}_{ELBO} = \alpha E_{q_\theta(z_x, z_{xy}^x | X_{t-1})} [\log p_\theta(\hat{X}_{t-1} | z_x, z_{xy}^x)] + \alpha E_{q_{\theta'}(z_y, z_{xy}^y | Y_t)} [\log p_{\theta'}(\hat{Y}_t | z_y, z_{xy}^y)] - D_{KL}(q || p),$$

where α is a weight paramter, and $D_{KL}(q || p)$ is the Kullback-Leibler divergence regularization term which regulates the variational posterior to follow the prior distribution.

Finally, the objective of CIC is to minimize the total loss, which is the weighted average of the losses $\mathcal{L}_{VAE}$ ($-\mathcal{L}_{ELBO}$), $\mathcal{L}_{diff}$ and $\mathcal{L}_{equal}$, i.e.,

$$\mathcal{L}_{loss} = \mathcal{L}_{VAE} + \beta_1 \mathcal{L}_{diff} + \beta_2 \mathcal{L}_{equal}, \quad (4.5)$$

where β_1 is the orthogonal loss coefficient and β_2 is the equivalent loss coefficient.

The CIC Algorithm is summarized in Algorithm 1. The implementation details, e.g., the selection and of VAE-related parameters and their impacts, are provided in **Supplementary Materials 1.5**.

4.5.3 Causality inference and latent confounders reconstruction by CIC

After the training of CIC with delay embedding vectors, the two latent spaces of X_{t-1} , i.e., z_{xy} and z_x generated by the encoder E_x , can be used to infer the causal interaction from x to y even under latent confounders according to Theorem 1 (Fig.S3). Note $CIC_{x \rightarrow y} \in [0, 1]$. Generally, CIC index is between 0 and 1 due to noise and distractions contained in real-world data. Therefore, thresholds, i.e. m and M , are selected to improve the robustness of CIC, in this work, m and M are set to 0.25 and 0.75, respectively. The details about the criteria for choosing the thresholds and the effects of different threshold settings are provided in Supplementary Materials 1.5 (4). Consequently, the CIC can be used to infer causality in the following way:

- (1) if $CIC_{x \rightarrow y} \in [0, m]$, then $x \nrightarrow y$;
- (2) if $CIC_{x \rightarrow y} \in (m, M)$, then $x \rightarrow y$ but with confounders
- (3) if $CIC_{x \rightarrow y} \in [M, 1]$, then $x \rightarrow y$.

The robustness and stability of the causal index were further validated under different perturbed parameters. As shown in Fig.

1D, the CIC score initially exhibited fluctuations but eventually stabilized to an asymptotic value as the perturbation parameters change. In addition, when $CIC_{x \rightarrow y} \geq M$, the higher the CIC score, the stronger the causality. In addition, in the CIC framework, the latent confounders can be quantified, i.e., as shown in Fig. 1D, the common information z_{xy} decomposed from x and y represents the latent confounder of x and y . To validate the consistency between the hidden variable z_{xy} and the real confounder Z , the Canonical Correlation Analysis (CCA) and the Mann-Whitney U Test were used to estimate the correlation and test the significance of distributional differences between z_{xy} and Z , respectively. The calculation details of CCA are as follows,

$$CCA(z_{xy}, z) = \sum_{i=1, j=1}^{n, m} \frac{Cov(z_{xy_i}, z_j)}{\sqrt{Var(z_{xy_i})Var(z_j)}},$$

where the n and m are the number of the samples of z_{xy} and z , respectively. z_{xy_i} is the i -th element of z_{xy} , and z_j is the j -th element of z .

The computational framework of CIC is summarized in Algorithm 1.

Algorithm 1: CIC

Input: The delay embedding variables X_{t-1} and Y_t ;

Params: $\alpha, \beta_1, \beta_2, l_{epoch}$;

CIC performs the following steps:

- 1: Randomly initialize the encoder parameters $\emptyset, \emptyset'$ and the decoder parameters θ, θ' .
- 2: **While** not reaching the maximal epochs l_{epoch} **do**
- 3: Calculate $\{\mu_{z_x}, \sigma_{z_x}\}, \{\mu_{z_{xy}^x}, \sigma_{z_{xy}^x}\}$ by encoder $\emptyset$ and $\{\mu_{z_y}, \sigma_{z_y}\}, \{\mu_{z_{xy}^y}, \sigma_{z_{xy}^y}\}$ by encoder $\emptyset'$.
- 4: Infer $q_\theta(z_x | X_{t-1})$, $q_\theta(z_{xy}^x | X_{t-1})$, $q_{\theta'}(z_y | Y_t)$ and $q_{\theta'}(z_{xy}^y | Y_t)$ by eqs (2.1~2.4) with reparameterization operations.
- 5: Reconstruct $p_\theta(X_{t-1} | \{z_x, z_{xy}^x\})$, $p_\theta(X_{t-1} | \{z_x^0, z_{xy}^x\})$ and $p_\theta(X_{t-1} | \{z_x, z_{xy}^0\})$ by decoder θ , and $p_{\theta'}(Y_t | \{z_y, z_{xy}^y\})$ by decoder θ' .
- 6: Update $\emptyset, \emptyset', \theta$ and θ' by minimize eq (2.5).
- 7: Compute the causal index

$$CIC_{x \rightarrow y} = \frac{MSE(\hat{X}^{z_x}, X)^2}{MSE(\hat{X}^{z_{xy}}, X)^2 + MSE(\hat{X}^{z_x}, X)^2}$$

based on the reconstructions $\hat{X}^{z_{xy}}$ and $\hat{X}^{z_x}$ by independent information z_x and common information z_{xy} , respectively.

8: end while

Output: The causal strength of computational version: $CIC_{x \rightarrow y}$.

5. RESULTS

In the following, we extensively validate the proposed CIC on various simulated and real-world datasets, including gene regulatory network, ecological network, and neuron network of rhesus monkey and rat circadian rhythm network. Firstly, the effectiveness and robustness of CIC are validated by three benchmark systems: a three-dimensional system with different

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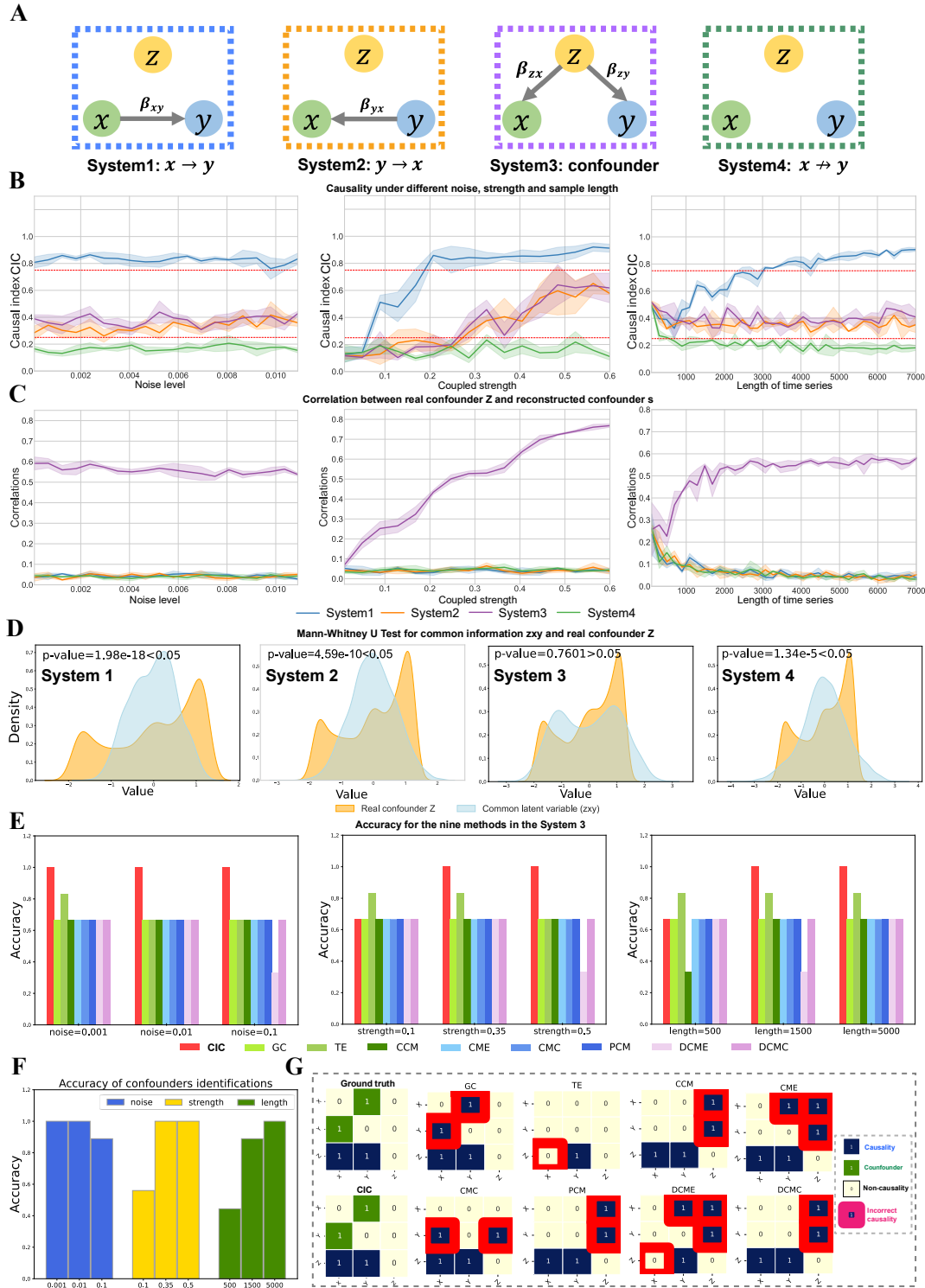


Fig. 2. Detecting causality in the three-dimensional system with different couplings. A. Four different logistic systems. System1: the causality from x to y ($\beta_{xy} = 0.35$, other coupled strengths=0), System2: the inverse causality from y to x ($\beta_{yx} = 0.35$), System3: the confounder of x and y ($\beta_{zx} = 0.35, \beta_{zy} = 0.35$), and System4: non-causality case (all the coupled strengths=0). B. The robustness and effectiveness of CIC varying noise levels, coupling strengths and length of time series. C. The correlations between true and reconstructed latent confounders (z_{xy}) under different parameters. D. The Mann-Whitney U Test for testing the significance of distributional differences between z_{xy} and Z . E. The comparison with other eight approaches for the System3. F. The accuracy of CIC in identifying confounders under different parameters. G. CIC vs. other dynamical methods (noise=0.001, causal strength=0.35, and length=5000). Color coding: blue for causality, green for confounders, yellow for non-causality, red for errors.

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couplings, an eight-dimensional logistic system, and the gene regulatory networks from DREAM4 *in silico* Network Challenge [66-68]. Then, four real-world datasets, including the food chain network [69-71], Hong Kong air pollution and cardiovascular disease dataset [72], Monkey brain neural dataset [14], and Rat circadian rhythm gene expression data [73] are used to validate the method. The sample sizes and numbers of variables of different datasets are provided in **Supplementary Materials 2.1**. Due to space limit, the results of the eight-dimensional logistic system and the Monkey brain neural dataset are provided in **Supplementary Materials 2.2-2.3**. We also evaluated the performance of CIC in some specific causal scenarios, including multiple-confounder scenarios, a mixture of direct causal effects and confounding effects, feedback loops and time-varying causal mechanisms (**Supplementary Materials 2.4**). We compare the CIC with eight other popular causal methods, including GC, TE, CCM, CME, CMC, PCM, DCME and DCMC. Specifically, GC, TE, CCM, CME, CMC infer causal interaction based on pairwise observed variables, similarly to CIC, while PCM, DCME, DCMC require additional observed variables as conditional variables to detect causal influence. As there is no universal standard to quantify the accuracy of causal inference, here we used the widely used indices to evaluate causal inference, e.g., AUC/ROC, Accuracy, Precision. In addition, since conventional methods cannot reconstruct latent confounders, then reconstructions of confounders by CIC are compared with the ground truth.

5.1 Inferring causality and latent confounders in three-dimensional discrete system

Consider a three-variable logistic system:

$$\begin{cases} x_{t+1} = x_t[\gamma_x - \gamma_x x_t - \beta_{yx} y_t - \beta_{zx} z_t] + \epsilon_{x,t} \\ y_{t+1} = y_t[\gamma_y - \gamma_y y_t - \beta_{xy} x_t - \beta_{zy} z_t] + \epsilon_{y,t} \\ z_{t+1} = z_t[\gamma_z - \gamma_z z_t] + \epsilon_{z,t} \end{cases}$$

where $\gamma_x = 3.7, \gamma_y = 3.72, \gamma_z = 3.78, \beta_{yx}, \beta_{zx}, \beta_{xy}, \beta_{zy}$ are coupling parameters, and different combinations of coupling parameters can lead to different network motifs. By setting different coupling parameters, we focus on the causal association from x to y under the four cases, as shown in **Fig. 2A**, including the causality from x to y , the inverse causality from y to x , the confounder of x and y , and non-causality between x and y . Then, we investigate the robustness and effectiveness of CIC at different parameter perturbation., e.g. different noises, coupling strength and length of time series. As shown in the left panel of **Fig. 2B**, CIC successfully infer the casual associations under various noise settings. Notably, CIC can distinguish direct cause $x \rightarrow y$ (System1) from the confounder case (System3) with sufficiently large coupled strength and a long length of the observed time series (**Fig. 2B**, right two panels). This capability is not achievable with other methods. **Fig. 2C** presents the latent confounder quantification for different parameters. Similarly, the high correlations indicates that CIC accurately reconstruct the latent confounder (**Fig. 2C**, the right two panels).

Specifically, in the first panel of **Fig. 2B-C**, the noise level is from 0.001 to 0.015, beyond which the logistic system will no longer converge. As the noise increases, the CIC can still

robustly infer causal interaction and eliminate non-causal cases. And for the System3 with a confounder, the correlation between the reconstructed confounder z_{xy} and the real confounder Z by CCA is asymptotically stabilized to 0.6, while the correlation values in other cases are close to 0 with noise fluctuating. In the second panel of **Fig. 2B-C**, the coupling strength is changed from 0 to 0.6, for which the CIC can also effectively infer causality and identify confounders as the strength increases, even under weak coupling or strong coupling. And the correlation between z_{xy} and Z stabilizes at 0.6. In the third panel of **Fig. 2B-C**, the length of time series varied from 100 to 10000. We can see that the CIC works well when the data length is greater than 2000. The above result is consistent with our theoretical results. **Fig. 2D** compares the distributions of z_{xy} and true confounder Z using Mann-Whitney U tests. The p values show the significance of distributional differences, for example, the p-values of 1.98e-18, 4.59e-10, 1.34e-5 (< 0.05) in system 1, 2 and 4 indicate the significant differences between common information z_{xy} and the true confounder Z , and the p-value of 0.7601 (> 0.05) shows that there is no statistically significant difference between z_{xy} and confounder Z , indicating the capability of CIC in reconstructing latent confounder.

To demonstrate robustness, we compared CIC with eight competing approaches, across variations in noise, causal strength and time-series length in System 3. **Fig. 2E** shows that CIC outperforms others in most scenarios, except at very weak causal strength ($\beta=0.1$) or short series (length=500), where TE performs better (the second panel). As causal strength increases, CIC's accuracy improves while TE's declines, with other methods showing lower overall accuracy and sensitivity. As the sample size increases, the performance of CIC surpasses existing methods, reaching an accuracy of 0.97. These results demonstrate that CIC's deep learning-based orthogonal decomposition significantly improves causal inference accuracy with unobserved confounders, though it requires adequate sample sizes. **Fig. 2G** shows that CIC can accurately infer all causal associations, but there are false positive results for GC, CCM, CME, CMC, PCM, DCMC and false negative situations for TE and DCME. In addition, CIC method is capable of identifying confounders, which is consistent with the ground truth. **Fig. 2F** illustrates the accuracy of CIC in identifying confounders under different parameters. We further verified the robustness and accuracy of CIC in more complex causal structures, e.g., multiple-confounder scenarios and the case of a mixture of direct causal effects and confounding effects. For the details, see **Supplementary Materials 2.4**.

5.2 Application of CIC in gene regulatory networks from DREAM4

We then applied CIC to infer gene regulatory networks (GRNs) collected from the DREAM4 *in silico* Network Challenge [66, 67]. There are ten GRNs with different structures, including five 10-node networks and five 100-node networks. The ground truth network was also distributed along the gene expression data, which allows us to assess the quality of inference result.

The inference result in causality and confounders by CIC and the comparison with other approaches are shown in **Fig. 3** and

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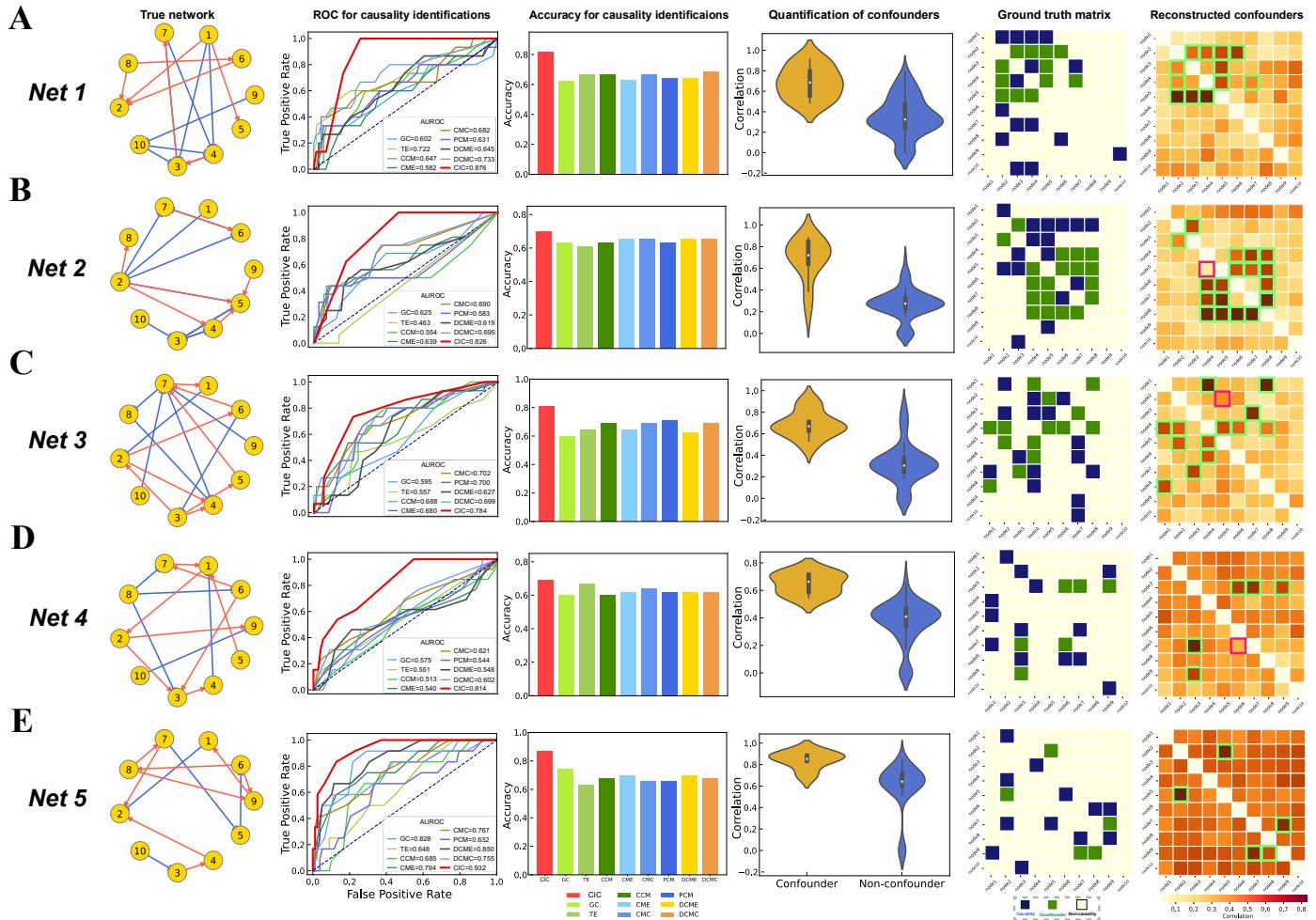


Fig. 3. Detecting causality in DREAM4 gene regulatory networks. A-E. The results of CIC in five 10-node networks. The first panel shows the true networks. The subsequent panels, from left to right, depict the comparative analysis between CIC and other methods using ROC curves, followed by bar plots representing accuracy metrics. The fourth panel presents the global correlation between the reconstructed confounders (common latent space) and the actual confounders. The fifth column depicts the ground truth matrix, with blue indicating causal links and green indicating confounders, and the sixth panel shows the reconstructed confounders from CIC by Heatmap.

Supplementary Materials 2.5. Fig. 3 illustrates the results of CIC method in five 10-node networks. Taking Net1 as an example, the first panel of **Fig. 3A** is the ground truth, and the second and third panels depict the results of the CIC and other methods. In the second and third panels, our method significantly outperforms other methods both by ROC and Accuracy criterion. Moreover, our method has the unique advantage of quantifying confounders. The fourth panel of **Fig. 3A** globally shows the correlation between the reconstructed confounders, i.e., the common latent variable, and the real confounders, which approaches to 0.7. While the correlation between the common latent variable of other non-confounder cases and the node with one of the lowest degrees in Net1, e.g., node 1, is less than 0.3. This shows that CIC can reconstruct latent confounders and avoid the misidentifications for the non-confounder cases. In addition, the details of the reconstructed confounder are further analyzed in the. The distribution of high correlations also reflects the consistency of the reconstructed confounders marked with green boxes in the sixth panel and the real confounders marked in the ground truth matrix from the fifth panel.

As shown in **Fig. 3B-3E**, CIC also perform better in Net 2-Net 4 than other eight dynamic causal methods.

The same analysis is also performed for the other five 100-nodes networks in DREAM4 dataset. The Accuracy criterion shows that CIC significantly exceeds other methods in the causality inference of high-dimensional gene regulatory networks. We also compare CIC with these four methods for reconstructing gene regulatory networks based on time-series data, e.g., such BETS [58], SWING [59] G1DBN [60] and DYNOTEARS[37]. The detailed result of five 100-node networks is provided in **Supplementary Materials 2.5**. CIC also accurately reconstructs latent confounders in the 100-node network, as shown in **Fig. S17**, panels 5-6. These results clearly validate the power of CIC to reconstruct confounders in a high-dimensional network.

5.3 Reconstruct food chain network of plankton species by CIC

We detected the causality in ecological network consisting of six creatures, i.e., Pico cyanobacteria, Rotifers, Nano flagellates,

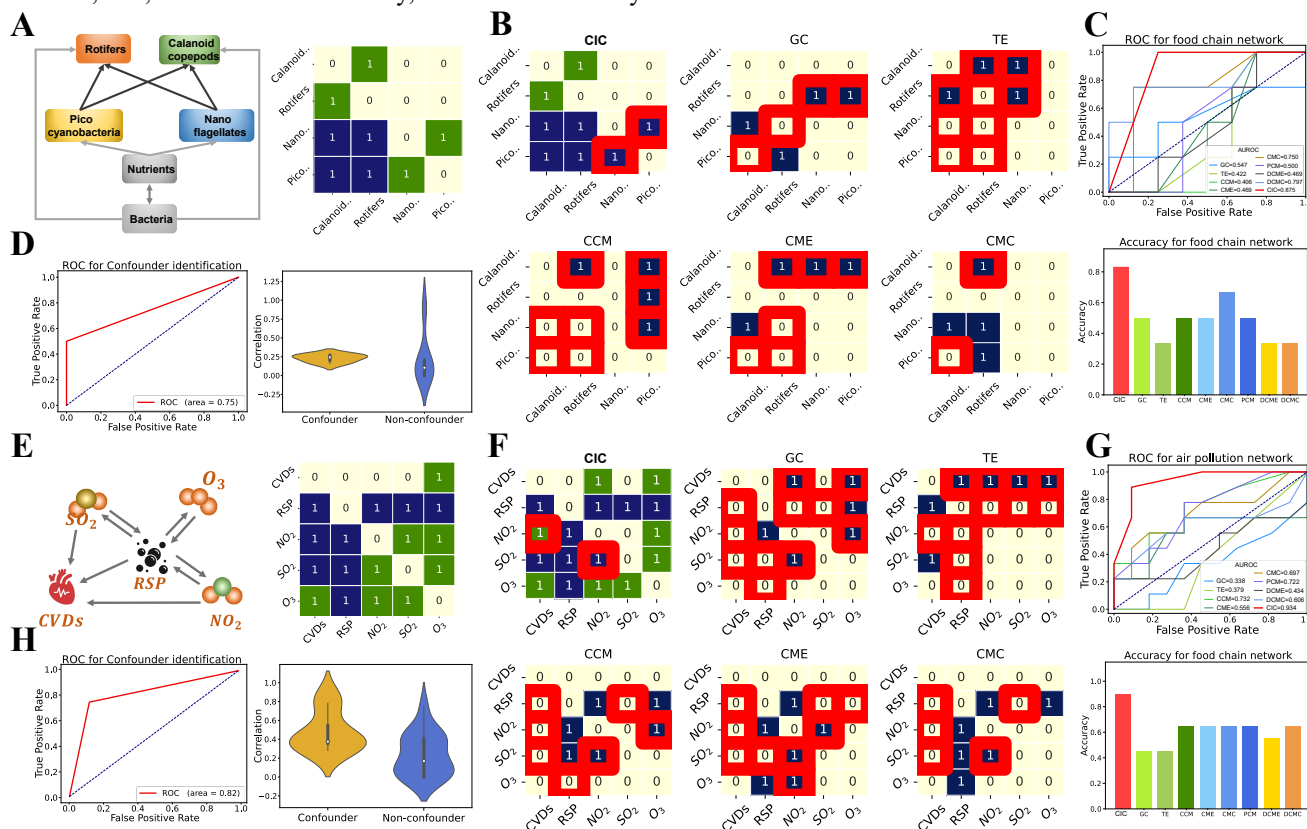
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Calanoid copepods, Bacteria and Nutrients. The first four form the observable food chain, while Bacteria and Nutrients are unobservable in this system. The oscillatory population data are selected from an 8-year mesocosm experiment of a plankton community isolated from the Baltic Sea [69-71], and the time series data with 794 time points were finally obtained after preprocessing. **Fig. 4A** illustrates the true food chain network. The second panel of **Fig. 4A** shows that there are three confounders, i.e., Nano flagellates, Pico cyanobacteria and unobservable Bacteria of Rotifers and Calanoid copepods, and one other confounder Nutrients of Nano flagellates and Pico cyanobacteria. **Fig. 4B** shows the results of CIC and other five (based on 2 observed variables) causal methods, in which the red grids represent wrong results. The comparison with conditional causal methods (based on 3 observed variables) is shown in **Supplementary Materials 2.6**. CIC successfully detected all true causal links in ground truth, while GC, TE, CCM, and CME produced wrong causal links, which obviously violated the ecological rule. Among the methods, CMC performs relatively better but missed one causal link. In addition, **Fig. 4C** benchmarks method performance by ROC curve and Accuracy. CIC also successfully identifies the confounders, including Nano flagellates, Pico cyanobacteria and Bacteria. However, the confounder effect caused by Nutrients are misidentified as a causal link, probably because of the strong noises of the real data. **Fig. 4D** shows that the correlations between true (Nano flagellates, Pico cyanobacteria) and reconstructed confounder, are above 0.25, and the correlations for the non-confounder case are close to 0.10. These correlations are overall lower than previous systems, probably due to the unknown observation data of a primary true confounder, i.e., Bacteria. In summary, the CIC accurately

inferred the causal links, identified latent confounders except Bacteria and effectively quantifies these confounders, in this real-world system.

5.4 Causality inference in Hong Kong cardiovascular disease and air pollution network

Strong associations between individual air pollutants and cardiovascular diseases (CVDs) have been found in previous study [72]. The data of concentrations of air pollutants and hospital admission of cardiovascular diseases in Hong Kong from 1994 to 1997 are used to infer the causal interactions in this work. The result is shown in **Fig. 4E-H**. **Fig. 4E** describes the true network among the pollutants and CVDs, SO₂, O₃, NO₂, respirable suspended particulates (RSP), and CVDs, and the second panel shows the detailed causalities and confounders. The first panel of **Fig. 4F** shows the inference result of CIC, i.e., the RSP, SO₂ are the cause of CVDs, which is consistent with previous studies [13, 74, 75]. In addition, a unidirectional causal interaction from SO₂ to NO₂ is detected, which is also found in previous study [3]. However, the causality from NO₂ to CVDs is not detected from the available data, as the causal link is regarded as the confounder effect of NO₂ and CVDs by CIC. It cannot be excluded other factors that are not completely available, such as temperature, air humidity, etc. By contrast, other methods lead to at least four false positive links when setting the 65% quantile [13, 14] of all causal strengths as the threshold value, such as the causal links from CVDs to NO₂ by GC and the causal links from CVDs to RSP by TE, which obviously violates the ecological rule. And other methods, e.g., CCM, CME and CMC cannot effectively infer the causality from pollutants to CVD, which may be affected by the strong noises of observed data or latent confounders.



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Fig. 4. Detecting causal links in two real-world networks. Food chain network of plankton species: **A.** The first panel is the food chain network. The second graph shows the causality and confounder in network. **B.** The inference result of CIC and other five (based on 2 observed variables) causal methods, with red grids indicating incorrect predictions. **C.** Comparison of all methods using ROC curves and accuracy metrics. **D.** Identification and quantification of confounders in food chain network. **Hong Kong air pollution and cardiovascular disease:** **E.** True network of Hong Kong air pollution and cardiovascular disease: the first graph shows causal links among pollutants SO₂, O₃, NO₂, RSP, and CVDs; The second shows the ground truth matrix with blue and green grids marking causal links and confounders. **F.** The inference result of CIC and other five (2-variable) methods in air pollution and cardiovascular disease network. **G.** The comparison of these methods based on ROC curve and Accuracy. **H.** The identification and quantification of confounders.

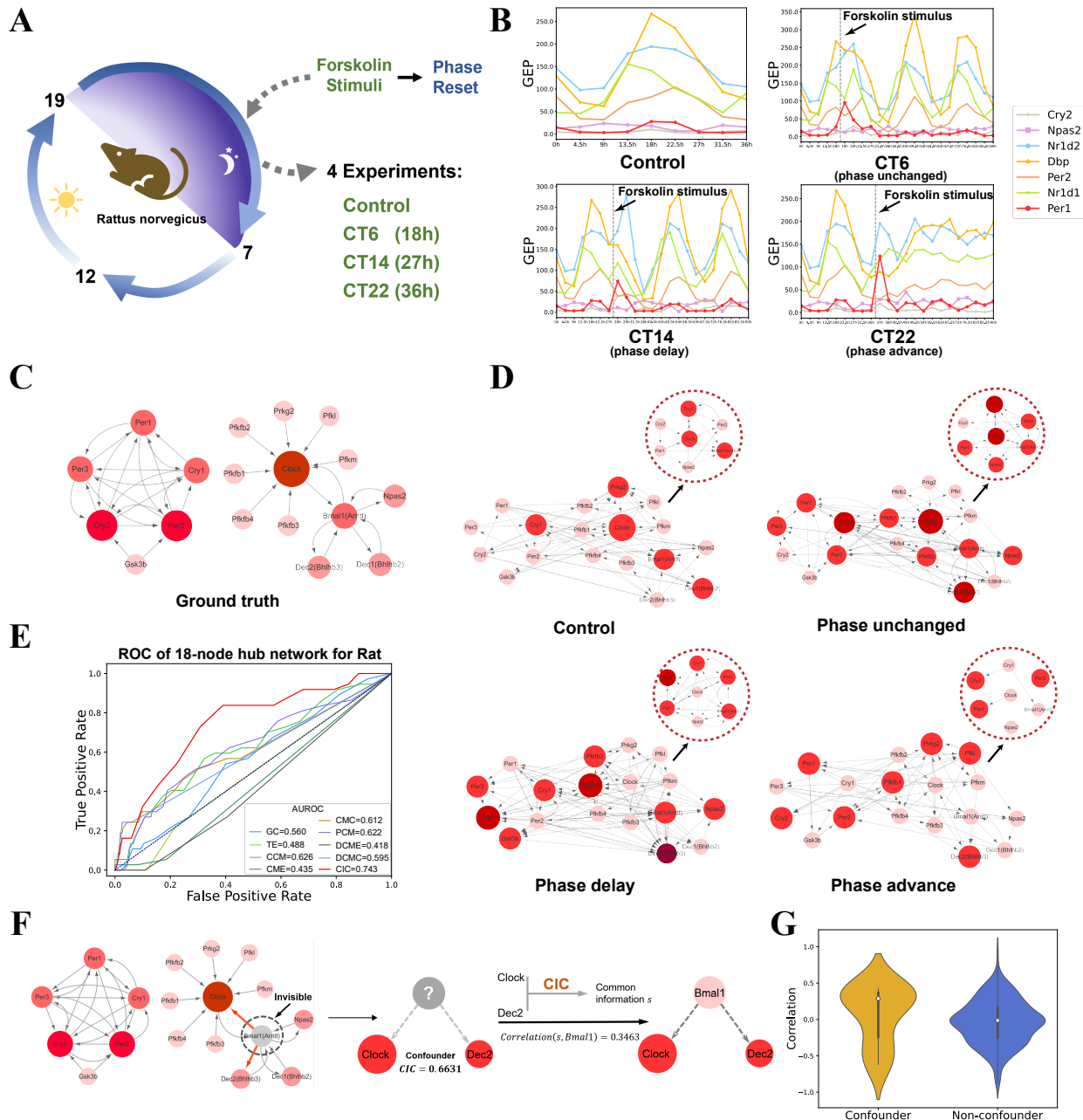


Fig. 5. Reconstructing the rat circadian rhythm network and latent confounder gene. **A.** Schematic diagram of the circadian rhythm experiments. **B.** Oscillation patterns of key circadian genes across four experiments. **C.** Ground truth of the 18-gene circadian rhythm network. **D.** The inferred circadian network for the four stage, control, phase unchanged, phase delay and phase advance. Hub sub-networks of seven key circadian genes are shown in the top right. **E.** ROC curves comparing CIC with eight other methods for the control experiment. **F.** To uncover hidden confounders, gene *Bmal1* (which regulates *Clock* and *Dec2*) is removed, and CIC is used to infer confounding structure between *Clock* and *Dec2*. **G.** the performance of our method in quantifying confounders of the circadian rhythm network.

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Fig. 4G provides the comparison between CIC and the above methods (based on two observed variables), and the comparison with 3-variable models is shown in **Supplementary Materials 2.6**. The result shows that our method outperforms other methods significantly by ROC curve and accuracy. In addition, the confounders are also effectively identified by CIC, such as the confounder of SO₂ and O₃, the confounder of CVDs and O₃, etc. **Fig. 4H** shows the identification and quantification of confounders, the correlation between the reconstructed one and the true confounders approaches to ~ 0.5 , and the average correlation of the non-confounder case is 0.2, indicating the power of our method in revealing the causal mechanism of the system.

5.5 Reconstruct Rat circadian rhythm network and latent gene by CIC

The circadian rhythm system is responsible for regulating various physiological and behavioral rhythms [76]. Extensive studies have revealed that a set of key circadian genes are capable of generating circadian oscillations [73]. However, how these circadian genes drive the circadian oscillation of gene expressions remain unknown. Thus, reconstructing the network of regulatory interactions among these circadian genes from a causal perspective is vital for understanding the processes that underlie circadian rhythms. Here, we used the CIC to a rat circadian rhythm gene expression data to detect causal interactions among genes to reconstruct the circadian network. The dataset includes four experiments (**Fig. 5A**), i.e. one control and three phase-shifted experiments. We provided the details about the dataset in **Supplementary Materials 2.1**.

Fig. 5B illustrates the dynamics of gene expression for the seven key circadian genes in the four experiments. The circadian oscillation of *Per1* expression can be used to identify the immediate phase-shifted by forskolin, and the master clock gene *Per1* shows phase unchanged in the CT6 experiment, phase delay in the CT14 experiment, and phase advance in the CT22 experiment (**Fig. 5B**). Then, we use the CIC algorithm to elucidate the gene regulatory network in the four different experiments. Here we consider a core regulatory network consisting of 18 genes, in which the transcriptional circuits are formed by regulation of E/E' boxes, DBP/E4BP4 binding elements, and RevErbA/ROR binding elements, respectively [77, 78]. **Fig. 5C** illustrates the ground truth of the 18-node circadian rhythm network. **Fig. 5D** shows the reconstructed circadian network for the four stage, control, phase unchanged, phase delay and phase advance, and the hub networks of the seven key circadian genes, e.g. *Cry1*, *Cry2*, *Per1*, *Per2*, *Clock*, *Npas2* and *Bmal1*(*Arntl*), are provided in the top right of **Fig. 5C**. It is clear that *Clock* and *Cry2* are two key genes that distinguish phase shifts. In the control and CT6 experiment, *Clock* gene is especially active to regulate other genes and *Cry2* is inactive with the lower degree. In the CT14 and CT22 experiments, the *Clock* gene becomes inactive with less causal links and the *Cry2* begins to play a core role in the network. Especially, at the CT6 experiments, after the stimulus by forskolin, the genes, *Per1*, *Per2* and *Npas2* actively participate in the regulation of the circadian rhythm network with dense causal links. In addition, for the phase delay stage in the CT14 experiment, the interactions of circadian rhythm network

increase and the regulation effects of the hub network are denser than the control, such as the degree of *Per1* and *Per2* both increases from 4 (of control) to 8. For the phase advance in the CT22 experiment, regulations of the circadian network become weaker and there are very sparse associations for the hub network. In addition, **Fig. 5E** further shows the ROC curve of CIC and the other eight compared causal methods for the circadian network reconstructions, CIC performs better for the inference of real circadian rhythm network with an AUROC of 0.743. **Fig. 5F-G** further demonstrates the potential of the method in identifying and reconstructing confounders. In **Fig. 5F**, there is a gene *Bmal1* which regulates the genes *Clock* and *Dec2*, then we removed the gene *Bmal1* from the 18-node circadian network, i.e. it becomes a latent confounder for *Clock* and *Dec2*, and calculated the CIC index to detect the causal interaction between *Clock* and *Dec2*. The result of the CIC score of 0.6631 shows that there are confounders of *Clock* and *Dec2* according to Theorem 1. And the correlation between the common information of *Clock* and *Dec2* (disentangled by CIC) and the real confounder *Bmal1*, is 0.3463. In addition, we also reconstruct other confounders of the network in the same way, as shown in **Fig. 5G**, the average correlation between the reconstructed confounders and real confounders is 0.35, while the correlations for the cases without confounders approaches 0, indicating the quality of latent confounders reconstruction by CIC. The low correlation values of 0.35 between the reconstructed confounders and real confounders may be influenced by other latent confounders outside of the 18-node circadian network. In summary, these above results demonstrate the ability of our method in reconstructing circadian rhythm network.

IV. DISCUSSION

We present CIC, a novel method to infer dynamical causality even with latent confounders. Our approach builds upon the Orthogonal Decomposition Theorem, implemented via a VAE framework for time-series data. The CIC method is applicable to time series data in either a linear or a nonlinear dynamical system even if there are latent confounders. CIC solves a long-standing challenge by enabling causal detection in high-dimensional systems using only two observed variables—regardless of unobserved confounders. Beyond its theoretical rigor, CIC offers two key advances: 1) detecting causal interactions with latent confounders using the data of only two observed variables and 2) reconstructing the latent confounders. Besides the latent confounder problem, our CIC can handle another notorious non-separability problem in causal inference.

Current approaches, including prediction-based methods and cross-mapping-based methods, face two key challenges: misidentifying the spurious interactions induced by confounders as causal relationships or suffering from reduced accuracy in the presence of multiple latent confounders. CIC overcomes these challenges by distinguishing genuine causality from confounder effects using just pairwise observations, without requiring additional variables. The method's core innovation is to apply orthogonal decomposition in delay embedding space, which transforms original non-separable dynamics into separable ones, thereby solving both the confounder challenge and non-separability problems

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simultaneously. In addition, to distinguish direct and indirect causality and identify visible and latent confounders, we extend Theorem 1 to a conditional version involving multiple observed variables, as outlined in Theorem 2 of **Supplementary Materials 1.3**. In addition, we adopt deep neural networks to efficiently realize the orthogonal decomposition to reliably detect causal links merely from observed time-series data even with unobserved confounders. The method is also applied to various benchmark and real-world datasets, demonstrating the power for inferring causality and latent confounders not only in a low-dimension system but also in a high-dimension system. Explicitly inferring the causalities and confounders is the key to recover the causal mechanisms of a complex system. Therefore, the CIC provides a powerful tool along this feat and effectively solves the confounder problem arising from previous influential research.

It is worth noting that there remain several issues that merit further investigation. Firstly, the orthogonal decomposition theorem is applicable to nonlinear dynamical systems with time-invariant assumption and the Takens' embedding theorem assumption, which limit the ability of CIC to directly capture time-varying causal mechanisms. Importantly, a practical extension of CIC to time-varying scenarios can be realized by combining CIC with a sliding-window approach (see **Supplementary Materials 2.4**). Secondly, when the observed correlation is a mixture of direct causal effects and confounding effects, our method can correctly identify the direct causal link but cannot accurately separate these two types of effects. In such cases, CIC prioritizes inferring the true causal relationship, meaning that the confounder alongside a true causal link does not affect the accuracy of causal structure, but may disturb the strength of causal interaction, leading to a relatively higher causal strength. However, increasing the sample size can help mitigate the confounding effects, making the estimated causal effects closer to the true causal effects (See **Supplementary Materials 2.4** for details). Thirdly, while CIC achieves superior accuracy through pairwise contrastive learning, its computational cost scales quadratically with the number of variables. However, the pairwise nature of CIC allows for massive parallelization, enabling scalability to larger systems via distributed computing clusters. For extremely high-dimensional systems (e.g., >1000 variables), we recommend a two-stage strategy: first applying a computationally efficient screening method to identify candidate pairs, and subsequently employing CIC for fine-grained causal verification. This hybrid approach balances scalability with the rigorous inference capabilities of CIC. Moreover, although modeling multiple latent confounders collectively as a latent variable z_{xy} is sufficient for eliminating spurious causality caused by latent confounders, disentangling specific confounding factors remains an exciting direction for future work. Finally, our study primarily focuses on dynamical causality within a non-intervention system. Addressing how to infer dynamical causality with interventions based solely on observed time series remains an important and intriguing problem for future research.

REFERENCES

- [1] A. Ma, X. Wang, J. Li *et al.*, "Single-cell biological network inference using a heterogeneous graph transformer," *Nature Communications*, vol. 14, no. 1, p. 964, 2023/02/21 2023, doi: 10.1038/s41467-023-36559-0.
- [2] E. Saberski, T. Lorimer, D. Carpenter *et al.*, "The impact of data resolution on dynamic causal inference in multiscale ecological networks," *Communications Biology*, vol. 7, no. 1, p. 1442, 2024/11/05 2024, doi: 10.1038/s42003-024-07054-z.
- [3] S. Leng, H. Ma, J. Kurths *et al.*, "Partial cross mapping eliminates indirect causal influences," *Nat Commun*, vol. 11, no. 1, p. 2632, May 26 2020, doi: 10.1038/s41467-020-16238-0.
- [4] G. Sugihara, R. May, H. Ye *et al.*, "Detecting causality in complex ecosystems," *Science*, vol. 338, no. 6106, pp. 496-500, Oct 26 2012, doi: 10.1126/science.1227079.
- [5] K. D. Hoover, "Causality in Economics and Econometrics," in *The New Palgrave Dictionary of Economics*. London: Palgrave Macmillan UK, 2017, pp. 1-13.
- [6] S. M. Hill, L. M. Heiser, T. Cokelaer *et al.*, "Inferring causal molecular networks: empirical assessment through a community-based effort," *Nat. Methods*, vol. 13, no. 4, pp. 310-318, 2016/04/01 2016, doi: 10.1038/nmeth.3773.
- [7] M. Kretschmer, D. Coumou, J. F. Donges, and J. Runge, "Using Causal Effect Networks to Analyze Different Arctic Drivers of Midlatitude Winter Circulation," (in English), *J. Clim.*, vol. 29, no. 11, pp. 4069-4081, 01 Jun. 2016 2016, doi: <https://doi.org/10.1175/JCLI-D-15-0654.1>.
- [8] S. Feuerriegel, D. Frauen, V. Melnychuk *et al.*, "Causal machine learning for predicting treatment outcomes," *Nat. Med.*, vol. 30, no. 4, pp. 958-968, 2024/04/01 2024, doi: 10.1038/s41591-024-02902-1.
- [9] K. Lagemann, C. Lagemann, B. Taschler, and S. Mukherjee, "Deep learning of causal structures in high dimensions under data limitations," *Nature Machine Intelligence*, vol. 5, no. 11, pp. 1306-1316, 2023/11/01 2023, doi: 10.1038/s42256-023-00744-z.
- [10] N. Fusi, O. Stegle, and N. Lawrence, "Accurate modeling of confounding variation in eQTL studies leads to a great increase in power to detect trans-regulatory effects," *Nature Precedings*, 2011/06/02 2011, doi: 10.1038/npre.2011.5995.1.
- [11] S. Zhao, W. Crouse, S. Qian, K. Luo, M. Stephens, and X. He, "Adjusting for genetic confounders in transcriptome-wide association studies improves discovery of risk genes of complex traits," *Nat. Genet.*, vol. 56, no. 2, pp. 336-347, 2024/02/01 2024, doi: 10.1038/s41588-023-01648-9.
- [12] L. Chen, C. Li, X. Shen, and W. Pan, "Discovery and inference of a causal network with hidden confounding," *arXiv e-prints*, p. arXiv:2309.09420, 2023, doi: 10.48550/arXiv.2309.09420.
- [13] J. Shi, L. Chen, and K. Aihara, "Embedding entropy: a nonlinear measure of dynamical causality," *J R Soc Interface*, vol. 19, no. 188, p. 20210766, Mar 2022, doi:

[1] A. Ma, X. Wang, J. Li *et al.*, "Single-cell biological

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

- 10.1098/rsif.2021.0766.
- [14] P. Tao, Q. Wang, J. Shi *et al.*, "Detecting dynamical causality by intersection cardinal concavity," *Fundamental Research*, 2023/02/06/ 2023, doi: 10.1016/j.fmre.2023.01.007.
- [15] D. Rubin, "Estimating causal effects of treatments in randomized and nonrandomized studies," *Journal of Educational Psychology*, vol. 66, 10/01 1974, doi: 10.1037/h0037350.
- [16] J. Pearl, "Causal Diagrams for Empirical Research," *Biometrika*, vol. 82, pp. 669-710, 12/01 1995, doi: 10.1093/biomet/82.4.669.
- [17] J. Pearl, "Causal Inference in Statistics: An Overview," *Statistics Surveys*, vol. 3, pp. 96-146, 01/01 2009, doi: 10.1214/09-SS057.
- [18] J. Pearl and D. Mackenzie, *The Book of Why: The New Science of Cause and Effect*. Basic Books, Inc., 2018.
- [19] P. Spirtes and C. Glymour, "An Algorithm for Fast Recovery of Sparse Causal Graphs," *Social Science Computer Review*, vol. 9, pp. 62 - 72, 1991.
- [20] S. Shimizu, P. Hoyer, A. Hyvärinen, and A. Kerminen, "A Linear Non-Gaussian Acyclic Model for Causal Discovery," *Journal of Machine Learning Research*, vol. 7, pp. 2003-2030, 10/01 2006.
- [21] Y. Chang and J. Dy, "Informative Subspace Learning for Counterfactual Inference," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 31, no. 1, 02/13 2017, doi: 10.1609/aaai.v31i1.10800.
- [22] K. Kuang, P. Cui, B. Li, M. Jiang, and S. Yang, "Estimating Treatment Effect in the Wild via Differentiated Confounder Balancing," presented at the Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Halifax, NS, Canada, 2017. [Online]. Available: <https://doi.org/10.1145/3097983.3098032>.
- [23] V. Del Tatto, G. Fortunato, D. Bueti, and A. Laio, "Robust inference of causality in high-dimensional dynamical processes from the Information Imbalance of distance ranks," *Proceedings of the National Academy of Sciences*, vol. 121, no. 19, p. e2317256121, 2024, doi: 10.1073/pnas.2317256121.
- [24] C. W. J. Granger, "Investigating Causal Relations by Econometric Models and Cross-spectral Methods," *Econometrica*, vol. 37, no. 3, pp. 424-38, 02/01 1969, doi: 10.2307/1912791.
- [25] S. Schiff, P. So, T. Chang, R. Burke, and T. Sauer, "Detecting dynamical interdependence and generalized synchrony through mutual prediction in a neural ensemble," *Physical review. E, Statistical physics, plasmas, fluids, and related interdisciplinary topics*, vol. 54, pp. 6708-6724, 01/01 1997, doi: 10.1103/PhysRevE.54.6708.
- [26] J. Arnhold, P. Grassberger, K. Lehnertz, and C. Elger, "A Robust Method for Detecting Interdependences: Application to Intracranially Recorded EEG," *Physica D: Nonlinear Phenomena*, vol. 134, pp. 419-430, 07/06 1999, doi: 10.1016/S0167-2789(99)00140-2.
- [27] T. Schreiber, "Measuring information transfer," *Phys Rev Lett*, vol. 85, no. 2, pp. 461-4, Jul 10 2000, doi: 10.1103/PhysRevLett.85.461.
- [28] H. Ma, S. Leng, C. Tao *et al.*, "Detection of time delays and directional interactions based on time series from complex dynamical systems," *Phys Rev E*, vol. 96, no. 1-1, p. 012221, Jul 2017, doi: 10.1103/PhysRevE.96.012221.
- [29] H. Ma, K. Aihara, and L. Chen, "Detecting causality from nonlinear dynamics with short-term time series," *Sci Rep*, vol. 4, p. 7464, Dec 12 2014, doi: 10.1038/srep07464.
- [30] H.-F. Ma, S. Leng, and L. Chen, "Detecting High-Dimensional Causal Networks using Randomly Conditioned Granger Causality," *CSIAM Transactions on Applied Mathematics*, vol. 2, no. 4, pp. 680-696, 12/01 2021, doi: 10.4208/csiam-am.2020-0184.
- [31] L. Yang, W. Lin, and S. Leng, "Conditional cross-map-based technique: From pairwise dynamical causality to causal network reconstruction," *Chaos*, vol. 33, no. 6, Jun 1 2023, doi: 10.1063/5.0144310.
- [32] A. Tank, I. Covert, N. Foti, A. Shojaie, and E. B. Fox, "Neural Granger Causality," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4267-4279, 2022, doi: 10.1109/tpami.2021.3065601.
- [33] Z. Yin and P. Barucca, "Deep recurrent modelling of Granger causality with latent confounding," *Expert Syst. Appl.*, vol. 207, no. C, p. 6, 2022, doi: 10.1016/j.eswa.2022.118036.
- [34] Z. Zhang, S. Ren, X. Qian, and N. Duffield, "InvarGC: Invariant Granger Causality for Heterogeneous Interventional Time Series under Latent Confounding," p. arXiv:2510.19138doi: 10.48550/arXiv.2510.19138.
- [35] J. Runge, A. Gerhardus, G. Varando, V. Eyring, and G. Camps-Valls, "Causal inference for time series," *Nature Reviews Earth & Environment*, vol. 4, no. 7, pp. 487-505, 2023/07/01 2023, doi: 10.1038/s43017-023-00431-y.
- [36] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, and D. Sejdinovic, "Detecting and quantifying causal associations in large nonlinear time series datasets," *Science Advances*, vol. 5, no. 11, p. eaau4996, 2019, doi: 10.1126/sciadv.aau4996.
- [37] R. Pamfil, N. Sriwattanaworachai, S. Desai *et al.*, "DYNOTEARS: Structure Learning from Time-Series Data," presented at the Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, Proceedings of Machine Learning Research, 2020. [Online]. Available: <https://proceedings.mlr.press/v108/pamfil20a.html>.
- [38] V. Chernozhukov, D. Chetverikov, M. Demirer *et al.*, "Double/debiased machine learning for treatment and structural parameters," *The Econometrics Journal*, vol. 21, no. 1, pp. C1-C68, 2018, doi: 10.1111/ectj.12097.
- [39] A. Santos, D. Rente, R. Seabra, and J. M. F. Moura, "Learning the causal structure of networked dynamical systems under latent nodes and structured noise," presented at the Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

- Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, 2024. [Online]. Available: <https://doi.org/10.1609/aaai.v38i13.29406>.
- [40] L. Cheng, R. Guo, and H. Liu, "Causal Mediation Analysis with Hidden Confounders," presented at the Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining, Virtual Event, AZ, USA, 2022. [Online]. Available: <https://doi.org/10.1145/3488560.3498407>.
- [41] P. Spirtes, C. Glymour, and R. Scheines, *Causation, Prediction, and Search*: The MIT Press, 2001. [Online]. Available: <https://doi.org/10.7551/mitpress/1754.001.0001>.
- [42] C. Diego, H. M. Marloes, K. Markus, and S. R. Thomas, "Learning high-dimensional directed acyclic graphs with latent and selection variables," *The Annals of Statistics*, vol. 40, no. 1, pp. 294-321, 2/1 2012, doi: 10.1214/11-AOS940.
- [43] C. Papagiannopoulou, D. Miralles, N. Verhoest, W. Dorigo, and W. Waegeman, "A non-linear Granger causality framework to investigate climate-vegetation dynamics," *Geoscientific Model Development Discussions*, pp. 1-24, 11/16 2016, doi: 10.5194/gmd-2016-266.
- [44] R. K. A. Karlsson and J. H. Krijthe, "Detecting hidden confounding in observational data using multiple environments," presented at the Proceedings of the 37th International Conference on Neural Information Processing Systems, New Orleans, LA, USA, 2023.
- [45] M. Negro, A. Piras, R. Ahsan, D. Arbour, and E. Zheleva, "Relational causal discovery with latent confounders," presented at the Proceedings of the Forty-First Conference on Uncertainty in Artificial Intelligence, Rio de Janeiro, Brazil, 2025.
- [46] D. M. Chickering, "Optimal structure identification with greedy search," *J. Mach. Learn. Res.*, vol. 3, no. null, pp. 507-554, 2003, doi: 10.1162/153244303321897717.
- [47] S. Zhu, I. Ng, and Z. Chen, "Causal Discovery with Reinforcement Learning," p. arXiv:1906.04477doi: 10.48550/arXiv.1906.04477.
- [48] B. Frot, P. Nandy, and M. H. Maathuis, "Robust Causal Structure Learning with Some Hidden Variables," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 81, no. 3, pp. 459-487, 2019, doi: 10.1111/rssb.12315.
- [49] R. Cai, Z. Huang, W. Chen, Z. Hao, and K. Zhang, "Causal discovery with latent confounders based on higher-order cumulants," presented at the Proceedings of the 40th International Conference on Machine Learning, Honolulu, Hawaii, USA, 2023.
- [50] D. Tramontano, Y. Kivva, S. Salehkaleybar, M. Drton, and N. Kiyavash, "Causal Effect Identification in lvLiNGAM from Higher-Order Cumulants," p. arXiv:2506.05202doi: 10.48550/arXiv.2506.05202.
- [51] C. Li, X. Shen, and W. Pan, "Nonlinear Causal Discovery with Confounders," p. arXiv:2302.03178doi: 10.48550/arXiv.2302.03178.
- [52] X. Zheng, B. Aragam, P. Ravikumar, and E. P. Xing, "DAGs with NO TEARS: continuous optimization for structure learning," presented at the Proceedings of the 32nd International Conference on Neural Information Processing Systems, Montréal, Canada, 2018.
- [53] Y. Yu, J. Chen, T. Gao, and M. Yu, "DAG-GNN: DAG Structure Learning with Graph Neural Networks," presented at the Proceedings of the 36th International Conference on Machine Learning, Proceedings of Machine Learning Research, 2019. [Online]. Available: <https://proceedings.mlr.press/v97/yu19a.html>.
- [54] J. M. Mooij, S. Magliacane, and T. Claassen, "Joint causal inference from multiple contexts," *J. Mach. Learn. Res.*, vol. 21, no. 1, p. Article 99, 2020.
- [55] G. Stepaniants, B. W. Brunton, and J. N. Kutz, "Inferring causal networks of dynamical systems through transient dynamics and perturbation," *Physical Review E*, vol. 102, no. 4, p. 042309, 10/26/2020, doi: 10.1103/PhysRevE.102.042309.
- [56] S. H. Park, S. Ha, and J. K. Kim, "A general model-based causal inference method overcomes the curse of synchrony and indirect effect," *Nature Communications*, vol. 14, no. 1, p. 4287, 2023/07/24 2023, doi: 10.1038/s41467-023-39983-4.
- [57] M. C. McGraw and E. A. Barnes, "Memory Matters: A Case for Granger Causality in Climate Variability Studies," (in English), *J. Clim.*, vol. 31, no. 8, pp. 3289-3300, 15 Apr. 2018 2018, doi: <https://doi.org/10.1175/JCLI-D-17-0334.1>.
- [58] J. Lu, B. Dumitrascu, I. C. McDowell *et al.*, "Causal network inference from gene transcriptional time-series response to glucocorticoids," *PLoS Comp. Biol.*, vol. 17, no. 1, p. e1008223, 2021, doi: 10.1371/journal.pcbi.1008223.
- [59] J. D. Finkle, J. J. Wu, and N. Bagheri, "Windowed Granger causal inference strategy improves discovery of gene regulatory networks," *Proceedings of the National Academy of Sciences*, vol. 115, no. 9, pp. 2252-2257, 2018, doi: doi:10.1073/pnas.1710936115.
- [60] S. Lèbre, "Inferring Dynamic Genetic Networks with Low Order Independencies," *Statistical Applications in Genetics and Molecular Biology*, vol. 8, no. 1, 2009, doi: doi:10.2202/1544-6115.1294.
- [61] J. Zhao, Z. Gan, R. Huang, C. Guan, J. Shi, and S. Leng, "Detecting dynamical causality via intervened reservoir computing," *Communications Physics*, vol. 7, no. 1, p. 232, 2024/07/12 2024, doi: 10.1038/s42005-024-01730-6.
- [62] A. Gerhardus and J. Runge, "High-recall causal discovery for autocorrelated time series with latent confounders," presented at the Proceedings of the 34th International Conference on Neural Information Processing Systems, Vancouver, BC, Canada, 2020.
- [63] M. Rolínek, D. Zietlow, and G. Martius, "Variational Autoencoders Pursue PCA Directions (by Accident)," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15-20 June 2019

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

- 2019, pp. 12398-12407, doi: 10.1109/CVPR.2019.01269.
- [64] A. Kumar and B. Poole, "On implicit regularization in β -VAEs," presented at the Proceedings of the 37th International Conference on Machine Learning, 2020.
- [65] C. Allen, "Unpicking Data at the Seams: Understanding Disentanglement in VAEs," 2024.
- [66] D. Marbach, T. Schaffter, C. Mattiussi, and D. Floreano, "Generating realistic in silico gene networks for performance assessment of reverse engineering methods," *J. Comput. Biol.*, vol. 16, no. 2, pp. 229-39, Feb 2009, doi: 10.1089/cmb.2008.09TT.
- [67] D. Marbach, R. J. Prill, T. Schaffter, C. Mattiussi, D. Floreano, and G. Stolovitzky, "Revealing strengths and weaknesses of methods for gene network inference," *Proc Natl Acad Sci U S A*, vol. 107, no. 14, pp. 6286-91, Apr 6 2010, doi: 10.1073/pnas.0913357107.
- [68] R. J. Prill, D. Marbach, J. Saez-Rodriguez *et al.*, "Towards a rigorous assessment of systems biology models: the DREAM3 challenges," *PLoS One*, vol. 5, no. 2, p. e9202, Feb 23 2010, doi: 10.1371/journal.pone.0009202.
- [69] E. Beninca, K. D. Johnk, R. Heerkloss, and J. Huisman, "Coupled predator-prey oscillations in a chaotic food web," *Ecol. Lett.*, vol. 12, no. 12, pp. 1367-78, Dec 2009, doi: 10.1111/j.1461-0248.2009.01391.x.
- [70] E. Beninca, J. Huisman, R. Heerkloss *et al.*, "Chaos in a long-term experiment with a plankton community," *Nature*, vol. 451, no. 7180, pp. 822-5, Feb 14 2008, doi: 10.1038/nature06512.
- [71] A. M. Neutel, J. A. Heesterbeek, and P. C. De Ruiter, "Stability in real food webs: weak links in long loops," *Science*, vol. 296, no. 5570, pp. 1120-3, May 10 2002, doi: 10.1126/science.1068326.
- [72] T. W. Wong, T. S. Lau, T. S. Yu *et al.*, "Air pollution and hospital admissions for respiratory and cardiovascular diseases in Hong Kong," *Occup. Environ. Med.*, vol. 56, no. 10, pp. 679-83, Oct 1999, doi: 10.1136/oem.56.10.679.
- [73] Y. Wang, X.-S. Zhang, and L. Chen, "A network biology study on circadian rhythm by integrating various omics data," *Omics : a journal of integrative biology*, vol. 13 4, pp. 313-24, 2009.
- [74] H. Mustafic, P. Jabre, C. Caussin *et al.*, "Main air pollutants and myocardial infarction: a systematic review and meta-analysis," *JAMA*, vol. 307, no. 7, pp. 713-21, Feb 15 2012, doi: 10.1001/jama.2012.126.
- [75] A. S. Shah, J. P. Langrish, H. Nair *et al.*, "Global association of air pollution and heart failure: a systematic review and meta-analysis," *Lancet*, vol. 382, no. 9897, pp. 1039-48, Sep 21 2013, doi: 10.1016/S0140-6736(13)60898-3.
- [76] P. L. Lowrey and J. S. Takahashi, "Mammalian circadian biology: elucidating genome-wide levels of temporal organization," *Annu Rev Genomics Hum Genet.*, vol. 5, pp. 407-41, 2004, doi: 10.1146/annurev.genom.5.061903.175925.
- [77] H. R. Ueda, S. Hayashi, W. Chen *et al.*, "System-level

identification of transcriptional circuits underlying mammalian circadian clocks," *Nat. Genet.*, vol. 37, no. 2, pp. 187-92, Feb 2005, doi: 10.1038/ng1504.

- [78] C. H. Ko and J. S. Takahashi, "Molecular components of the mammalian circadian clock," *Hum. Mol. Genet.*, vol. 15 Spec No 2, pp. R271-7, Oct 15 2006, doi: 10.1093/hmg/ddl207.



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